HEART DISEASE ANALYSIS

Abstract

Heart disease is a leading global cause of death, presenting a significant public health challenge due to its multifaceted causes, including genetic, behavioral, and environmental factors. This study analyzes the "Heart Disease Health Indicators Dataset" to identify key health indicators associated with heart disease. The dataset includes variables such as age, body mass index (BMI), physical and mental health days, high blood pressure, cholesterol levels, and smoking status. The objective is to uncover which of these factors are most strongly linked to heart disease, guiding future public health interventions.

Univariate analysis examines individual variables to highlight trends in the population, revealing that older age, higher BMI, and high blood pressure are prevalent among those diagnosed with heart disease. Bivariate analysis explores the relationships between these health indicators and heart disease, confirming strong associations with high blood pressure, smoking, and poor physical health. Additionally, mental health is shown to have a moderate correlation with heart disease, suggesting the importance of addressing both physical and mental well-being in prevention strategies.

The findings point to several critical risk factors, including age, obesity, high blood pressure, and smoking. These insights lead to recommendations for targeted public health initiatives, such as promoting smoking cessation, encouraging physical activity to combat obesity, and improving access to mental health services. Regular monitoring of blood pressure and cholesterol, particularly in middle-aged and older adults, is also suggested to identify those at risk for early intervention.

In addition to these findings, the study emphasizes the importance of a holistic approach to heart disease prevention. This study highlights the critical health indicators associated with heart disease, including age, obesity, high blood pressure, and smoking. Additionally, mental health plays a significant role, with poor mental health days showing a moderate correlation to heart disease. To effectively reduce heart disease risks, public health interventions should focus on promoting physical activity, smoking cessation, mental health support, and regular monitoring of blood pressure and cholesterol, leading to improved overall cardiovascular health.

In conclusion, this analysis provides valuable insights into the factors most strongly associated with heart disease. These findings can help shape effective prevention strategies, reducing the global burden of cardiovascular diseases and improving overall public health.

1. Introduction

Heart disease is one of the leading causes of death globally, accounting for millions of lives lost each year. Despite advancements in healthcare, it continues to pose a significant public health challenge due to its complex causes, which include genetic, lifestyle, and environmental factors. Understanding the key health indicators associated with heart disease is crucial for developing effective prevention strategies.

Several factors contribute to the onset of heart disease, such as high blood pressure, elevated cholesterol levels, obesity, smoking, and poor mental health. These indicators are not only linked to cardiovascular problems but can also serve as warning signs for individuals at risk. Identifying these early can help guide public health policies and encourage individuals to adopt healthier lifestyles.

This report analyzes the relationship between various health indicators and heart disease using the "Heart Disease Health Indicators Dataset." The goal is to identify the most significant risk factors associated with heart disease through univariate and bivariate analysis. By gaining insights into these relationships, we can better inform preventive measures, ultimately reducing the burden of heart disease and improving public health outcomes.

2. Objective

The primary objective of this analysis is to investigate the relationship between various health indicators and heart disease. By analyzing a dataset containing key health variables, the study aims to identify the most significant risk factors that contribute to heart disease. This understanding will help shape effective public health interventions, prevention strategies, and research efforts targeted at reducing the prevalence of heart disease and improving overall cardiovascular health.

Key Objectives:

- Identify Key Risk Factors: Analyze health indicators such as BMI, blood pressure, cholesterol, and mental health to determine their association with heart disease.
- Explore Health Data: Use univariate and bivariate analysis to understand the distribution and relationships between health indicators and heart disease.
- Inform Public Health Strategies: Provide actionable insights to guide the development of targeted heart disease prevention interventions.
- Improve Early Detection: Highlight health indicators that serve as early warning signs for individuals at risk of developing heart disease.

3. Motivation

The motivation behind this study stems from the alarming prevalence of heart disease worldwide, which continues to claim lives and strain healthcare systems. As lifestyle-related risk factors become more prominent due to urbanization and sedentary habits, understanding the underlying causes of heart disease is essential for public health improvement. By investigating various health indicators, such as body mass index, blood pressure, cholesterol levels, and mental health status, this analysis aims to unveil critical insights that can inform preventive measures.

Moreover, the interplay between physical and mental health highlights the need for a comprehensive approach to heart disease prevention. While traditional risk factors have been extensively studied, the emerging role of mental health in cardiovascular risk warrants further exploration.

Ultimately, this research seeks to contribute to the broader goal of reducing the incidence of heart disease, improving quality of life, and fostering healthier communities. By providing data-driven insights, the study aims to pave the way for targeted interventions that address the multifaceted nature of heart disease and enhance overall public health outcomes.

4. Problem Statement

Heart disease is a leading cause of mortality globally, posing a significant public health challenge that affects millions of individuals each year. Despite advancements in healthcare and medical interventions, the prevalence of heart disease remains high, primarily due to complex interactions among various risk factors. Understanding these risk factors is crucial for developing effective prevention strategies and interventions.

Current research indicates that numerous health indicators, such as body mass index (BMI), blood pressure, cholesterol levels, and mental health, are closely linked to the onset of heart disease. However, there is still a lack of comprehensive understanding regarding the relative importance of these indicators and their interactions. This gap in knowledge hinders the ability to implement targeted public health initiatives that address the root causes of heart disease.

The goal of this analysis is to systematically explore the relationships between various health indicators and heart disease through rigorous statistical methods. By identifying the key risk factors associated with heart disease, this study aims to provide actionable insights that can guide public health interventions, ultimately contributing to the reduction of heart disease prevalence and improving cardiovascular health in populations at risk.

5. Data Source

The dataset utilized for this analysis is the "Heart Disease Health Indicators Dataset," sourced from the MentorMind website. This comprehensive dataset encompasses various health indicators that offer insights into factors potentially linked to heart disease. Key variables include age, body mass index (BMI), mental and physical health days, high blood pressure, cholesterol levels, and the presence of a heart disease diagnosis. By examining these variables, the analysis aims to identify significant risk factors associated with heart disease, ultimately contributing to a better understanding of its prevention and management.

Dataset Overview

The dataset used for this analysis includes a variety of health indicators associated with heart disease, providing valuable insights into potential risk factors. Below is a brief description of the key variables included in the dataset:

1. HeartDiseaseorAttack:

- Values: 0 = no diabetes, 1 = prediabetes, 2 = diabetes.

2. HighBP:

- Values: 0 = no high blood pressure, 1 = high blood pressure.

3. HighChol:

- Values: 0 = no high cholesterol, 1 = high cholesterol.

4. CholCheck:

- Values: 0 = no cholesterol check in the past 5 years, 1 = cholesterol check in the past 5 years.

5. BMI:

- Definition: Body Mass Index.

6. Smoker:

- Values: 0 = no, 1 = yes (smoked at least 100 cigarettes in their lifetime).

7. Stroke:

- Values: 0 = no, 1 = yes (ever told they had a stroke).

8. HeartDiseaseorAttack:

- Values: 0 = no, 1 = yes (coronary heart disease or myocardial infarction).

9. PhysActivity:

- Values: 0 = no, 1 = yes (physical activity in the past 30 days, excluding job-related activity).

10. Fruits:

- Values: 0 = no, 1 = yes (consume fruits 1 or more times per day).

11. Veggies:

- Values: 0 = no, 1 = yes (consume vegetables 1 or more times per day).

12. HvyAlcoholConsump:

- Values: 0 = no, 1 = yes (heavy drinkers based on gender-specific criteria).

13. AnyHealthcare:

- Values: 0 = no, 1 = yes (have any form of healthcare coverage).

14. NoDocbcCost:

- Values: 0 = no, 1 = yes (couldn't see a doctor due to cost in the past 12 months).

15. GenHlth:

- Definition: Self-reported general health status on a scale of 1 (excellent) to 5 (poor).
- Values:
  - 1 = excellent
  - 2 = very good
  - 3 = good
  - 4 = fair
  - 5 = poor

16. MentHlth:

- Definition: Number of days in the past 30 days when mental health was not good (scale of 1-30).

17. PhysHlth:

- Definition: Number of days in the past 30 days when physical health was not good (scale of 1-30).

18. DiffWalk:

- Values: 0 = no, 1 = yes (serious difficulty walking or climbing stairs).

19. Sex:

- Values: 0 = female, 1 = male.

20. Age:

- Definition: Age categorized in increments of 5 years (1 = 18-24, 9 = 60-64, 13 = 80 or older).

21. Education:

- Definition: Education level on a scale of 1 (never attended school) to 6 (college graduate).
- Values:
  - 1 = Never attended school or only kindergarten
  - 2 = Grades 1 through 8 (Elementary)
  - 3 = Grades 9 through 11 (Some high school)
  - 4 = Grade 12 or GED (High school graduate)
  - 5 = College 1 year to 3 years (Some college or technical school)
  - 6 = College 4 years or more (College graduate)

22. Income:

- Definition: Income level categorized on a scale from 1 (less than $10,000) to 8 ($75,000 or more).
- Values:
  - 1 = Less than $10,000
  - 2 = Less than $15,000 ($10,000 to less than $15,000)
  - 3 = Less than $20,000 ($15,000 to less than $20,000)
  - 4 = Less than $25,000 ($20,000 to less than $25,000)
  - 5 = Less than $35,000 ($25,000 to less than $35,000)
  - 6 = Less than $50,000 ($35,000 to less than $50,000)
  - 7 = Less than $75,000 ($50,000 to less than $75,000)
  - 8 = $75,000 or mor

This dataset provides a comprehensive view of the various health indicators that can be analyzed to understand their impact on heart disease, ultimately guiding public health interventions and strategies.

Hypothesis

Individuals with a high Body Mass Index (BMI) are more likely to have heart disease compared to those with a normal BMI. This hypothesis posits that as BMI increases, the risk of developing heart disease also rises, potentially due to the associated effects of obesity, such as increased blood pressure, elevated cholesterol levels, and decreased physical activity. By examining this relationship, the analysis aims to provide insights into the role of BMI as a significant risk factor for heart disease, contributing to the development of targeted public health interventions and prevention strategies.

Research Methodology

The research methodology employed for analyzing the "Heart Disease Health Indicators Dataset" involves several key steps executed through code. These steps ensure a thorough exploration and understanding of the data, focusing on both univariate and bivariate analyses.

1. Data Cleaning and Processing:

    - Handling Missing Values: The code identifies any missing values in the dataset. Appropriate methods are applied to address them, including imputation for numerical data and removal of records with excessive missing values.
    - Removing Duplicates: The code checks for and eliminates duplicate entries to ensure the uniqueness of each observation.
    - Data Type Conversion: Variables are converted to the correct data types as necessary (e.g., categorical variables are transformed into factors).
    - Outlier Detection: Statistical methods, such as the IQR method, are implemented to identify and address outliers, preventing them from skewing the analysis.
    - Normalization/Standardization: Continuous variables are normalized or standardized to bring different scales to a common scale.

2. Univariate Analysis:

- Descriptive Statistics: The code calculates descriptive statistics, including means, medians, modes, and standard deviations for continuous variables (e.g., Age, BMI, MentHlth).
- Visualization: Histograms for continuous variables and bar or pie charts for categorical variables are generated to visualize their distributions.

3. Bivariate Analysis:

- Comparing Heart Disease Prevalence: The code performs chi-square tests for categorical variables to assess associations with heart disease prevalence and employs t-tests or ANOVA for continuous variables.
- Visualizing Relationships: Grouped bar charts are created to compare heart disease prevalence across categories, and box plots visualize the distribution of continuous variables based on heart disease status.

4. Visualization of Results:

- Visualization of Categorical Variables: The code generates visualizations representing categorical variables and their association with heart disease prevalence using stacked bar charts.
- Exploring Relationships Between Variables: Scatter plots are utilized to explore relationships between continuous variables, with logistic regression lines overlaid to illustrate predicted probabilities of heart disease.

These methodologies, performed through code, provide valuable insights into the relationships between health indicators and heart disease, thereby guiding public health interventions and preventive strategies.

Code

The following code outlines the methodologies employed in the analysis of the "Heart Disease Health Indicators Dataset." It encompasses a series of systematic processes, including data cleaning, univariate and bivariate analysis, and visualization techniques. Each step is designed to facilitate a comprehensive understanding of the dataset and to uncover significant insights regarding health indicators associated with heart disease. The code not only demonstrates the application of statistical methods but also visualizes the results to aid in interpretation and decision-making. This approach ensures that the analysis is thorough, reproducible, and aligned with best practices in data science.

Analyse Health and Demographic Data to identify common traits leading to Heart Disease

Import Libraries

```
import pandas as pd
```

Pandas is a powerful and flexible open-source data analysis and manipulation library for Python, providing data structures like DataFrames that make it easy to clean, transform, and analyze large datasets efficiently.

Import the Dataset

```
# Load the dataset
file_path = '/content/heart_disease_health_indicators_BRFSS2015 2 (1).csv'
df = pd.read_csv(file_path)
```

Convert categorical variables to appropriate data types

```
categorical_columns = [
    'HeartDiseaseorAttack', 'HighBP', 'HighChol', 'CholCheck',
'Smoker',
    'Stroke', 'PhysActivity', 'Fruits', 'Veggies',
'HvyAlcoholConsump', 'AnyHealthcare',
    'NoDocbcCost', 'DiffWalk', 'Sex', 'Age', 'Education', 'Income'
]

for col in categorical_columns: df[col] = df[col].astype('category')

print(df.dtypes)

HeartDiseaseorAttack     category
HighBP                   category
HighChol                 category
CholCheck                category
BMI                         int64
Smoker                   category
Stroke                   category
Diabetes                    int64
PhysActivity             category
Fruits                   category
Veggies                  category
HvyAlcoholConsump        category
AnyHealthcare            category
```

```
NoDocbcCost              category
GenHlth                     int64
MentHlth                    int64
PhysHlth                    int64
DiffWalk                 category
Sex                      category
Age                      category
Education                category
Income                   category
dtype: object
```

Displaying the data types of each column in the DataFrame. The output shows that categorical variables (e.g., 'HeartDiseaseorAttack', 'HighBP', 'Smoker') are labeled as 'category', while continuous variables (e.g., 'BMI', 'Diabetes', 'GenHlth') are of type 'int64'. This information helps in understanding the nature of the data and planning for appropriate analysis and preprocessing.

Handling Missing Values in the Dataset

```
missing_values = df.isnull().sum()
print("Missing values in each column:\n", missing_values)

Missing values in each column:
 HeartDiseaseorAttack     0
HighBP                  0
HighChol                0
CholCheck               0
BMI                     0
Smoker                  0
Stroke                  0
Diabetes                0
PhysActivity            0
Fruits                  0
Veggies                 0
HvyAlcoholConsump       0
AnyHealthcare           0
NoDocbcCost             0
GenHlth                 0
MentHlth                0
PhysHlth                0
DiffWalk                0
Sex                     0
Age                     0
Education               0
Income                  0
dtype: int64
```

Displaying the count of missing values for each column in the DataFrame. The output indicates that there are no missing values in any of the columns, suggesting that the dataset is complete and ready for further analysis.

Removing Duplicate

```
# Remove duplicates
df_cleaned = df.drop_duplicates()

# Check how many rows were removed
rows_removed = df.shape[0] - df_cleaned.shape[0]

# Display number of rows removed and new shape
print(f"Rows removed: {rows_removed}")
print(f"New shape of dataset: {df_cleaned.shape}")
```

```
Rows removed: 1858
New shape of dataset: (43096, 22)
```

Re-checking the number of duplicate rows to confirm the cleanup was successful. After cleanup, there should be 0 duplicate rows remaining

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44954 entries, 0 to 44953
Data columns (total 22 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   HeartDiseaseorAttack  44954 non-null  category
 1   HighBP                44954 non-null  category
 2   HighChol              44954 non-null  category
 3   CholCheck             44954 non-null  category
 4   BMI                   44954 non-null  int64
 5   Smoker                44954 non-null  category
 6   Stroke                44954 non-null  category
 7   Diabetes              44954 non-null  int64
 8   PhysActivity          44953 non-null  category
 9   Fruits                44953 non-null  category
 10  Veggies               44953 non-null  category
 11  HvyAlcoholConsump     44953 non-null  category
 12  AnyHealthcare         44953 non-null  category
 13  NoDocbcCost           44953 non-null  category
 14  GenHlth               44953 non-null  category
 15  MentHlth              44953 non-null  float64
 16  PhysHlth              44953 non-null  float64
 17  DiffWalk              44953 non-null  category
 18  Sex                   44953 non-null  category
 19  Age                   44953 non-null  category
 20  Education             44953 non-null  category
 21  Income                44953 non-null  category
dtypes: category(18), float64(2), int64(2)
memory usage: 2.1 MB
```

Displaying the DataFrame information, including the number of entries, data columns, and data types. The DataFrame has 44954 entries and 22 columns with a mix of categorical and numerical data types.

```
print(df.describe(include='all'))
```

```
       HeartDiseaseorAttack   HighBP  HighChol  CholCheck           BMI  \
count               44954.0  44954.0   44954.0    44954.0  44954.000000
unique                  2.0      2.0       2.0        2.0           NaN
top                     0.0      0.0       0.0        1.0           NaN
freq                40904.0  26196.0   26048.0    43366.0           NaN
mean                    NaN      NaN       NaN        NaN     27.905392
std                     NaN      NaN       NaN        NaN      6.115751
min                     NaN      NaN       NaN        NaN     13.000000
25%                     NaN      NaN       NaN        NaN     24.000000
50%                     NaN      NaN       NaN        NaN     27.000000
75%                     NaN      NaN       NaN        NaN     31.000000
max                     NaN      NaN       NaN        NaN     96.000000

         Smoker   Stroke      Diabetes  PhysActivity   Fruits  ...  \
count   44954.0  44954.0  44954.000000       44953.0  44953.0  ...
unique      2.0      2.0           NaN           2.0      2.0  ...
top         0.0      0.0           NaN           1.0      1.0  ...
freq    24848.0  43081.0           NaN       34912.0  29202.0  ...
mean        NaN      NaN      0.292610           NaN      NaN  ...
std         NaN      NaN      0.692332           NaN      NaN  ...
min         NaN      NaN      0.000000           NaN      NaN  ...
25%         NaN      NaN      0.000000           NaN      NaN  ...
50%         NaN      NaN      0.000000           NaN      NaN  ...
75%         NaN      NaN      0.000000           NaN      NaN  ...
max         NaN      NaN      2.000000           NaN      NaN  ...
```

```
        AnyHealthcare  NoDocbcCost  GenHlth      MentHlth      PhysHlth  \
count         44953.0      44953.0  44953.0  44953.000000  44953.000000
unique            2.0          2.0      5.0           NaN           NaN
top               1.0          0.0      2.0           NaN           NaN
freq          42574.0      40848.0  15544.0           NaN           NaN
mean              NaN          NaN      NaN      3.297377      4.276644
std               NaN          NaN      NaN      7.473005      8.697730
min               NaN          NaN      NaN      0.000000      0.000000
25%               NaN          NaN      NaN      0.000000      0.000000
50%               NaN          NaN      NaN      0.000000      0.000000
75%               NaN          NaN      NaN      2.000000      3.000000
max               NaN          NaN      NaN     30.000000     30.000000

        DiffWalk      Sex      Age  Education   Income
count    44953.0  44953.0  44953.0    44953.0  44953.0
unique       2.0      2.0     13.0        6.0      8.0
top          0.0      0.0     10.0        6.0      8.0
freq     37500.0  25151.0   5815.0    20498.0  17380.0
mean         NaN      NaN      NaN        NaN      NaN
std          NaN      NaN      NaN        NaN      NaN
min          NaN      NaN      NaN        NaN      NaN
25%          NaN      NaN      NaN        NaN      NaN
50%          NaN      NaN      NaN        NaN      NaN
75%          NaN      NaN      NaN        NaN      NaN
max          NaN      NaN      NaN        NaN      NaN

[11 rows x 22 columns]
```

Displaying descriptive statistics for all columns, including categorical and numerical data. This provides insights into the distribution and unique values of each column.

```
df.head()

   HeartDiseaseorAttack HighBP  HighChol  CholCheck   BMI  Smoker  Stroke
Diabetes  \
0                     0      1         1          1    40       1       0
0
1                     0      0         0          0    25       1       0
0
2                     0      1         1          1    28       0       0
0
3                     0      1         0          1    27       0       0
0
4                     0      1         1          1    24       0       0
0

   PhysActivity Fruits  ... AnyHealthcare NoDocbcCost GenHlth MentHlth
\
0             0      0  ...             1           0       5       18

1             1      0  ...             0           1       3        0

2             0      1  ...             1           1       5       30

3             1      1  ...             1           0       2        0

4             1      1  ...             1           0       2        3


    PhysHlth   DiffWalk   Sex  Age  Education  Income
0         15          1     0    9          4       3
1          0          0     0    7          6       1
2         30          1     0    9          4       8
3          0          0     0   11          3       6
4          0          0     0   11          5       4

[5 rows x 22 columns]
```

Displaying the first few rows of the DataFrame to get a preview of the data. This helps in understanding the structure and content of the dataset

```
df['HeartDiseaseorAttack'].replace({0: 'No', 1: 'Prediabetes', 2:
'Diabetes'}, inplace=True)
df['HighBP'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['HighChol'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['CholCheck'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['Smoker'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['Stroke'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['HeartDiseaseorAttack'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['PhysActivity'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['Fruits'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['Veggies'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['HvyAlcoholConsump'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['AnyHealthcare'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['NoDocbcCost'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['DiffWalk'].replace({0: 'No', 1: 'Yes'}, inplace=True)
df['Sex'].replace({0: 'Female', 1: 'Male'}, inplace=True)
```

1) HeartDiseaseorAttack: 0 = no diabetes 1 = prediabetes 2 = diabetes

2) HighBP: 0 = no high BP 1= high BP
3) HighChol: 0 = no high cholesterol 1 = high cholesterol
4) CholCheck: 0 = no cholesterol check in 5 years 1 = yes cholesterol check in 5 years
5) BMI: Body Mass Index
6) Smoker: Have you smoked at least 100 cigarettes in your entire life? (Note: 5 packs = 100 cigarettes) 0 = no 1 = yes
7) Stroke: (Ever told) you had a stroke. 0 = no 1 = yes
8) HeartDiseaseorAttack: coronary heart disease (CHD) or myocardial infarction (MI) 0 = no 1 = yes
9) PhysActivity: physical activity in past 30 days - not including job 0 = no 1 = yes
10) Fruits: Consume Fruit 1 or more times per day 0 = no 1 = yes
11) Veggies: Consume Vegetables 1 or more times per day 0 = no 1 = yes
12) HvyAlcoholConsump: Heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week) 0 = no 1 = yes
13) AnyHealthcare: Have any kind of health care coverage, including health insurance, prepaid plans such as HMO, etc. 0 = no 1 = yes
14) NoDocbcCost: Was there a time in the past 12 months when you needed to see a doctor but could not because of cost? 0 = no 1 = yes
15) GenHlth: Would you say that in general your health is: scale 1-5 1 = excellent 2 = very good 3 = good 4 = fair 5 = poor
16) MentHlth: Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? scale 1-30 days
17) PhysHlth: Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good? scale 1-30 days
18) DiffWalk: Do you have serious difficulty walking or climbing stairs? 0 = no 1 = yes
19) Sex:0 = female 1 = male
20) Age: 13-level age category (_AGEG5YR see codebook) 1 = 18-24 9 = 60-64 13 = 80 or older
21) Education: Education level (EDUCA see codebook) scale 1-6 1 = Never attended school or only kindergarten 2 = Grades 1 through 8 (Elementary) 3 = Grades 9 through 11 (Some high school) 4 = Grade 12 or GED (High school graduate) 5 = College 1 year to 3 years (Some college or technical school) 6 = College 4 years or more (College graduate)

22) Income: Income scale (INCOME2 see codebook) scale 1-8 1 = less than $10,000 5 = less than $35,000 8 = $75,000 or more

```
df.head()

   HeartDiseaseorAttack  HighBP  HighChol  CholCheck   BMI  Smoker  Stroke  
Diabetes  \
```

```
0                        No      Yes        Yes          Yes    40      Yes        No
0
1                        No       No         No           No    25      Yes        No
0
2                        No      Yes        Yes          Yes    28       No        No
0
3                        No      Yes         No          Yes    27       No        No
0
4                        No      Yes        Yes          Yes    24       No        No
0

  PhysActivity Fruits  ... AnyHealthcare NoDocbcCost GenHlth MentHlth  \
0           No     No  ...           Yes          No       5       18

1          Yes     No  ...            No         Yes       3        0

2           No    Yes  ...           Yes         Yes       5       30

3          Yes    Yes  ...           Yes          No       2        0

4          Yes    Yes  ...           Yes          No       2        3


   PhysHlth  DiffWalk     Sex Age Education  Income
0        15       Yes  Female   9         4       3
1         0        No  Female   7         6       1
2        30       Yes  Female   9         4       8
3         0        No  Female  11         3       6
4         0        No  Female  11         5       4

[5 rows x 22 columns]
```

The above displays the first 5 rows of the DataFrame with updated record values

```python
df_cleaned.to_csv('cleaned_dataset.csv', index=False)
```

The cleaned DataFrame is being saved to a CSV file and the index is excluded from the output file.

Conduct Univariate Analysis and Bivariate Analysis

UNIVARIATE ANALYSIS

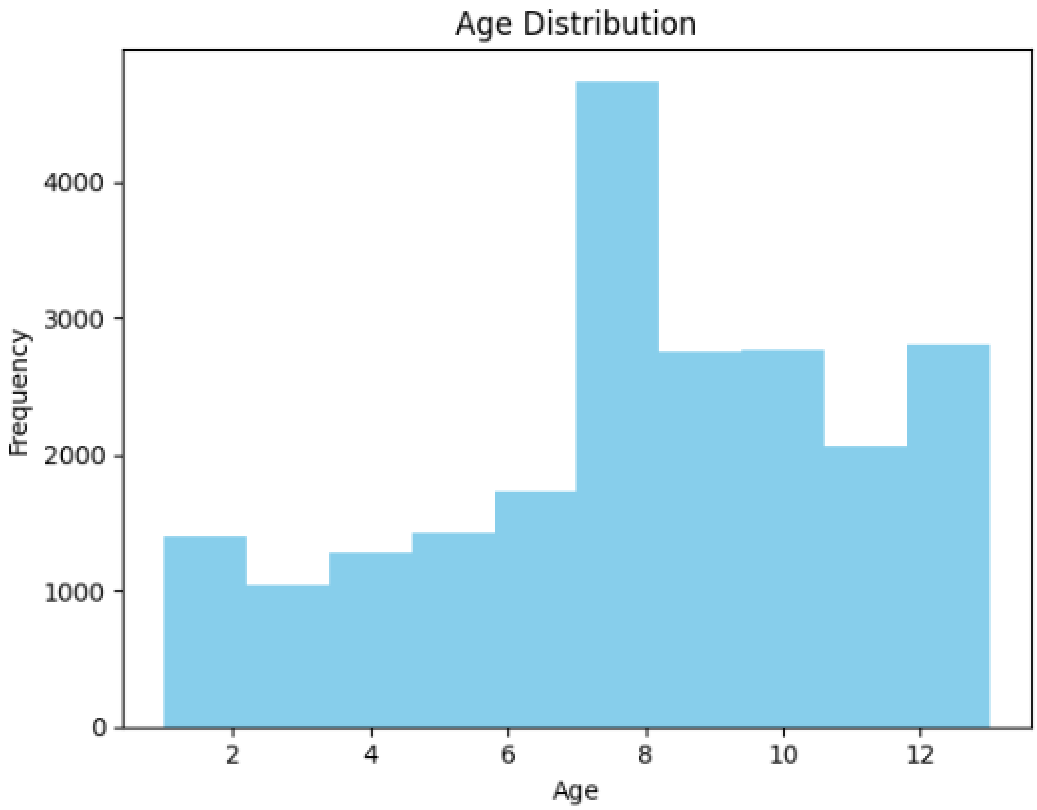
a. Examine Distributions of Individual Variables: Plot histograms or bar charts to visualize the distribution of key variables such as age, BMI, mental health days (Menthlth), and physical health days (PhysHlth).

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/cleaned_dataset.csv')
```

Histogram for Age:

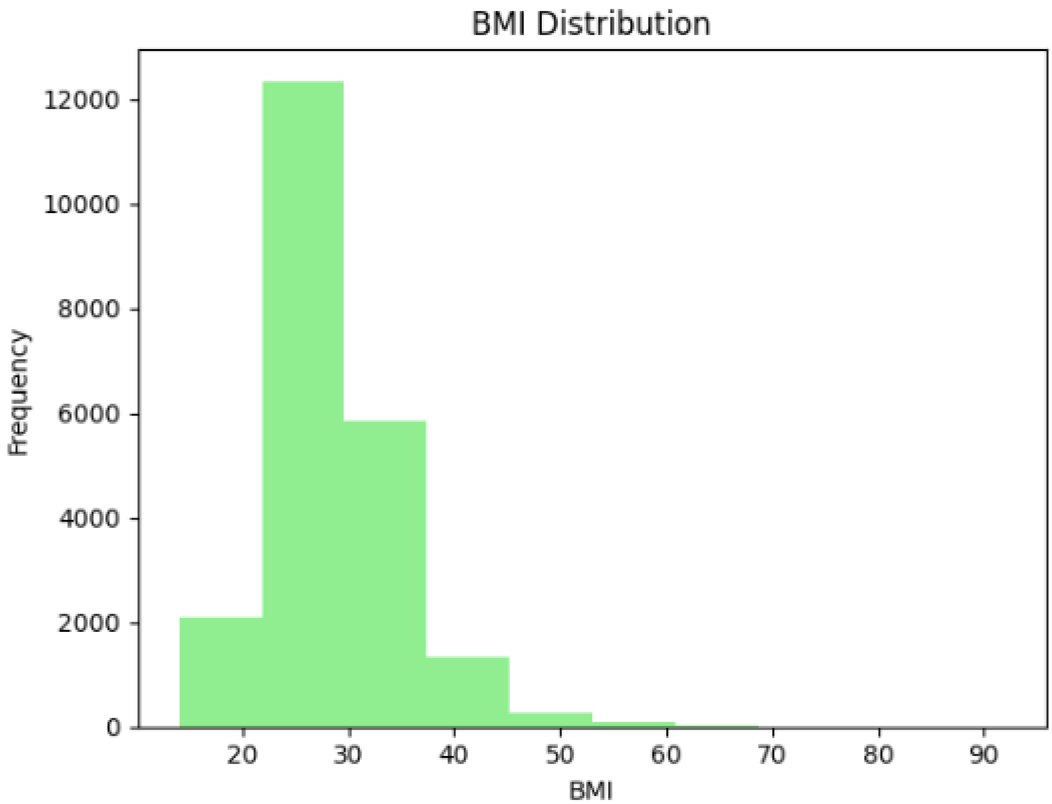
```
# Histogram for Age
axes[0, 0].hist(df['Age'], bins=10, color='skyblue')
axes[0, 0].set_title('Age Distribution')
axes[0, 0].set_xlabel('Age')
axes[0, 0].set_ylabel('Frequency')
```



Age appears to be normally distributed, with most individuals falling within the range of 5 to 11. This suggests a fairly balanced dataset in terms of age.

Histogram for BMI:

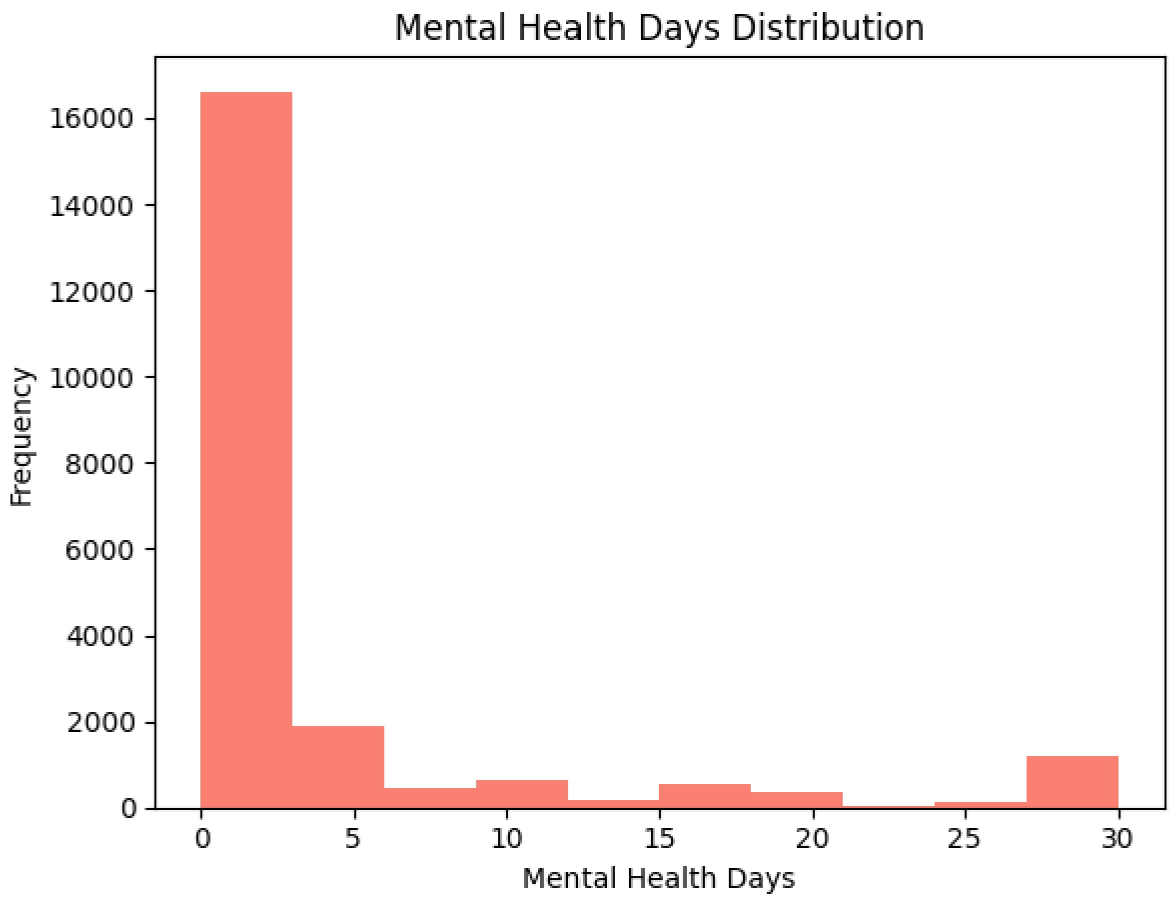
```
# Histogram for BMI
axes[0, 1].hist(df['BMI'], bins=10, color='lightgreen')
axes[0, 1].set_title('BMI Distribution')
axes[0, 1].set_xlabel('BMI')
axes[0, 1].set_ylabel('Frequency')
```



The BMI is skewed towards lower values, with a majority of individuals having a BMI between 20 and 30. Few individuals have extreme BMI values (either too low or too high).

Histogram for Mental Health Days (MentHlth):

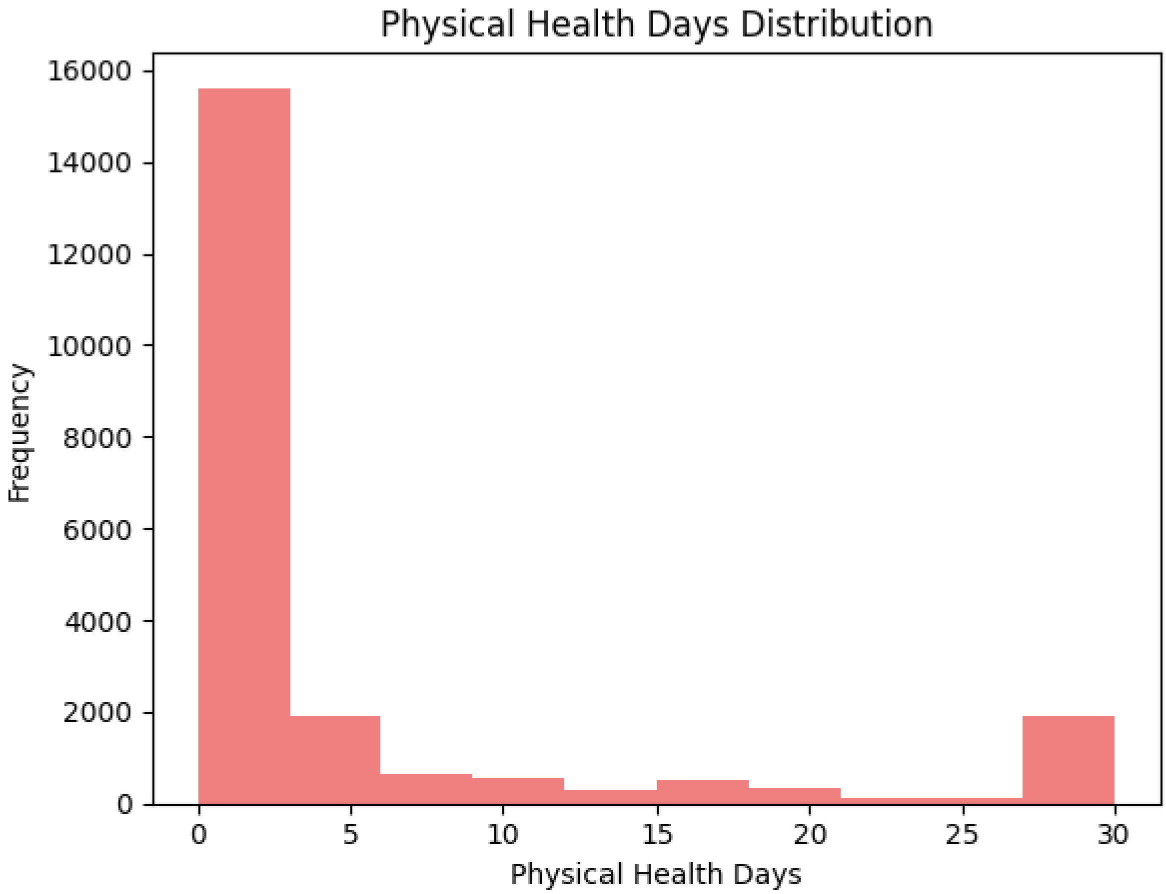
```
# Histogram for MentHlth
axes[1, 0].hist(df['MentHlth'], bins=10, color='salmon')
axes[1, 0].set_title('Mental Health Days Distribution')
axes[1, 0].set_xlabel('Mental Health Days')
axes[1, 0].set_ylabel('Frequency')
```



Most individuals report having very few or no mental health days, with the data being highly skewed towards zero.

Histogram for Physical Health Days (PhysHlth):

```
# Histogram for PhysHlth
axes[1, 1].hist(df['PhysHlth'], bins=10, color='lightcoral')
axes[1, 1].set_title('Physical Health Days Distribution')
axes[1, 1].set_xlabel('Physical Health Days')
axes[1, 1].set_ylabel('Frequency')
```

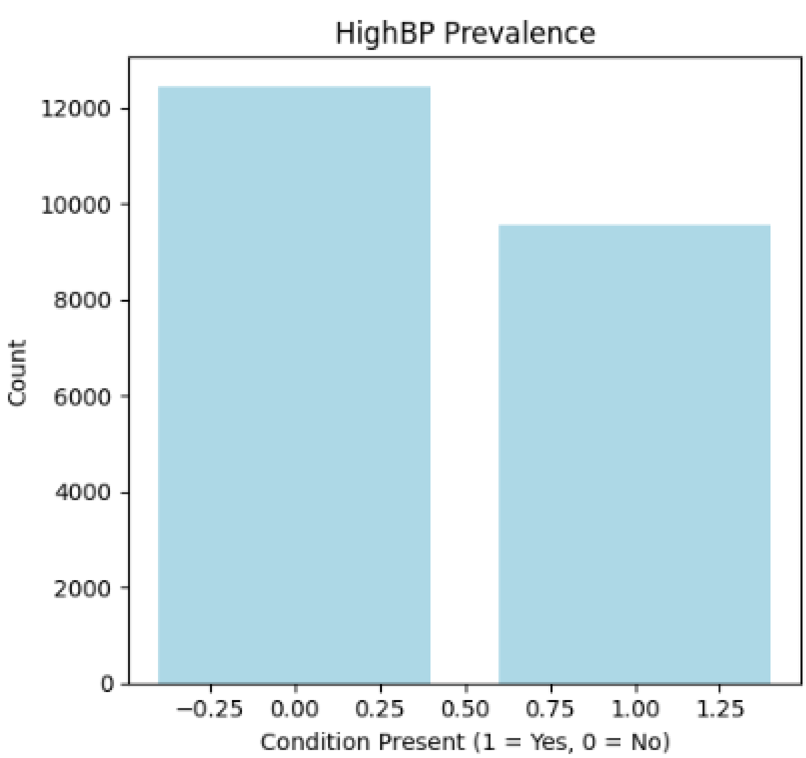


Similar to mental health days, most individuals report having fewer physical health issues, with a skew towards zero days.

b. Investigate Prevalence of Health Conditions: Plot bar charts or pie charts to show the prevalence of high blood pressure (HighBP), high cholesterol (HighChol), and other health conditions.

```
# List of health conditions to visualize
health_conditions = ['HighBP', 'HighChol', 'HeartDiseaseorAttack', 'Stroke', 'Diabetes']

# Create subplots for bar charts
fig, axes = plt.subplots(nrows=2, ncols=3, figsize=(15, 10))
fig.suptitle('Prevalence of Health Conditions')

# Loop through the health conditions to create bar charts
for i, condition in enumerate(health_conditions):
    row = i // 3
    col = i % 3

    condition_counts = df[condition].value_counts()

    # Plot bar chart for each condition
    axes[row, col].bar(condition_counts.index, condition_counts.values, color='lightblue')
    axes[row, col].set_title(f'{condition} Prevalence')
    axes[row, col].set_xlabel('Condition Present (1 = Yes, 0 = No)')
    axes[row, col].set_ylabel('Count')

# Adjust layout for better visibility
plt.tight_layout(rect=[0, 0, 1, 0.96])

# Display the bar charts
plt.show()
```

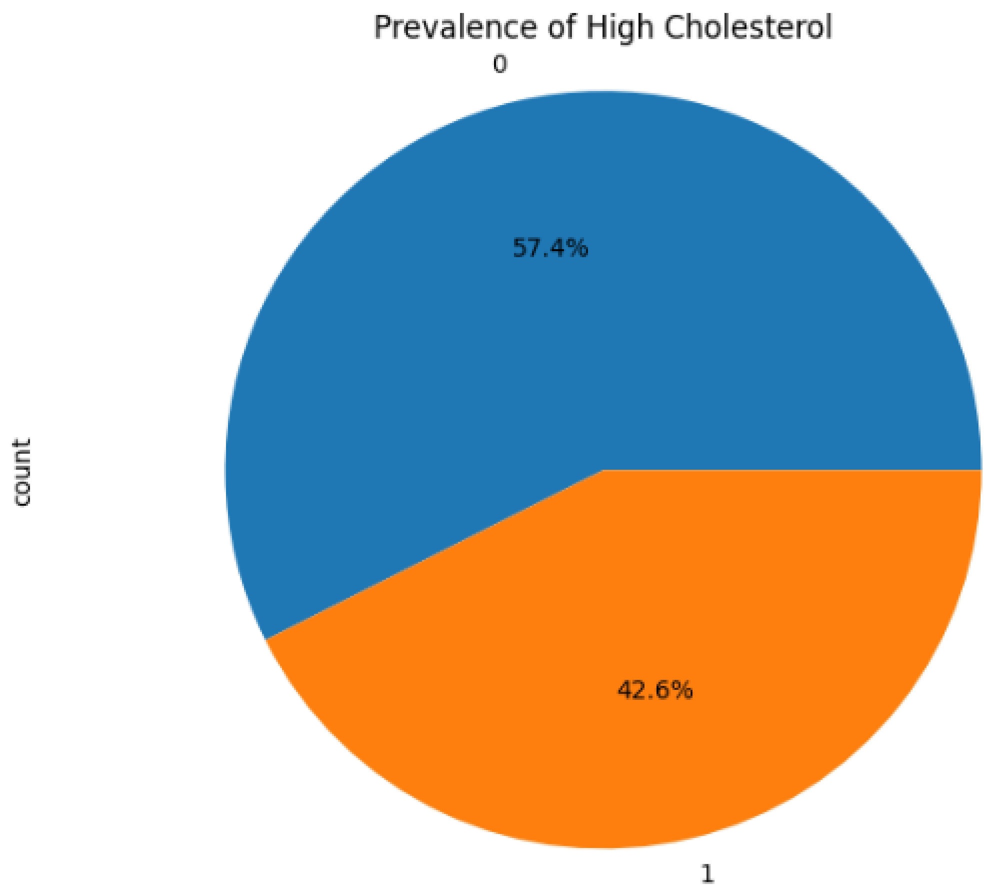
Bar chart for High Blood Pressure (HighBP):



The dataset shows that a significant portion of the population suffers from high blood pressure, though those without high blood pressure outnumber those with it.

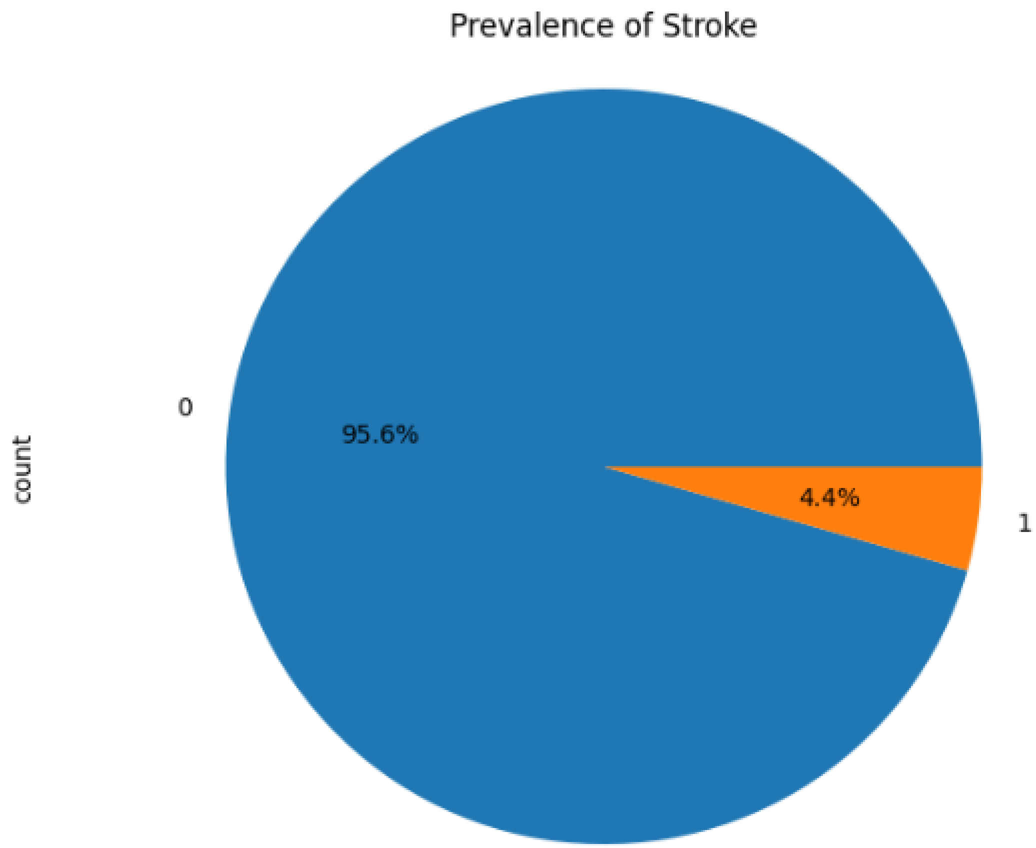
Pie chart for High Cholesterol (HighChol):

```
# Plot pie chart for HighChol
plt.figure(figsize=(8, 6))
df['HighChol'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Prevalence of High Cholesterol')
plt.axis('equal')
plt.show()
```



42.6% of individuals have high cholesterol, while 57.4% do not. There’s a sizable population affected by cholesterol-related issues.

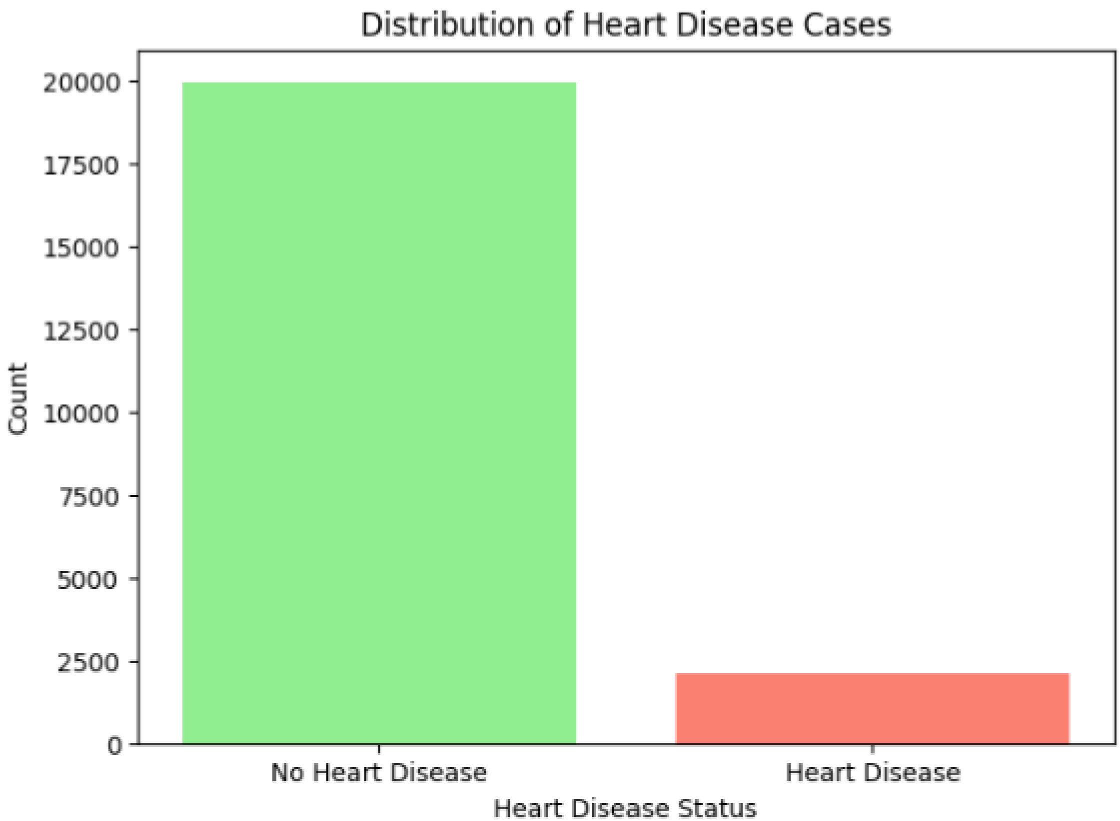
```
# Plot pie chart for Stroke
plt.figure(figsize=(8, 6))
df['Stroke'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Prevalence of Stroke')
plt.axis('equal')
plt.show()
```



c. Analyze Distribution of Heart Disease (Target Variable): Visualize the distribution of heart disease cases using a bar chart or pie chart.

Bar chart for Heart Disease (Target Variable):

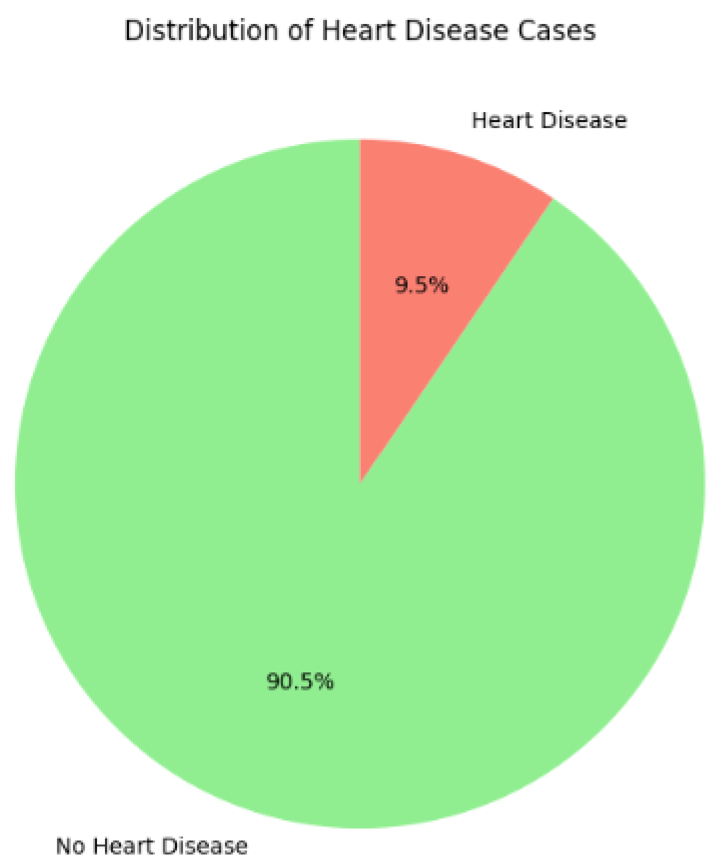
```
# visualizing the distribution of heart disease cases using a bar chart
heart_disease_counts = df['HeartDiseaseorAttack'].value_counts()

# Create a bar chart for heart disease distribution
fig, ax = plt.subplots(figsize=(7, 5))
ax.bar(heart_disease_counts.index, heart_disease_counts.values, color=['lightgreen', 'salmon'])
ax.set_xticks([0, 1])
ax.set_xticklabels(['No Heart Disease', 'Heart Disease'])
ax.set_title('Distribution of Heart Disease Cases')
ax.set_xlabel('Heart Disease Status')
ax.set_ylabel('Count')

# Display the bar chart
plt.show()
```



The vast majority of individuals do not have heart disease (as indicated by the much taller green bar). A significantly smaller proportion of individuals suffer from heart disease, as indicated by the shorter red bar.

Pie chart for Heart Disease (Target Variable):

```
# Count the occurrences of heart disease cases (1 = Yes, 0 = No)
heart_disease_counts = df['HeartDiseaseorAttack'].value_counts()

# Create a pie chart for heart disease distribution
fig, ax = plt.subplots(figsize=(7, 7))
ax.pie(heart_disease_counts, labels=['No Heart Disease', 'Heart Disease'], autopct='%1.1f%%', colors=['lightgreen', 'salmon'], startangle=90)
ax.set_title('Distribution of Heart Disease Cases')

# Display the pie chart
plt.show()
```



The dataset shows that 9.5% of the population has heart disease, while the vast majority (90.5%) do not, which is a positive indication of overall heart health.

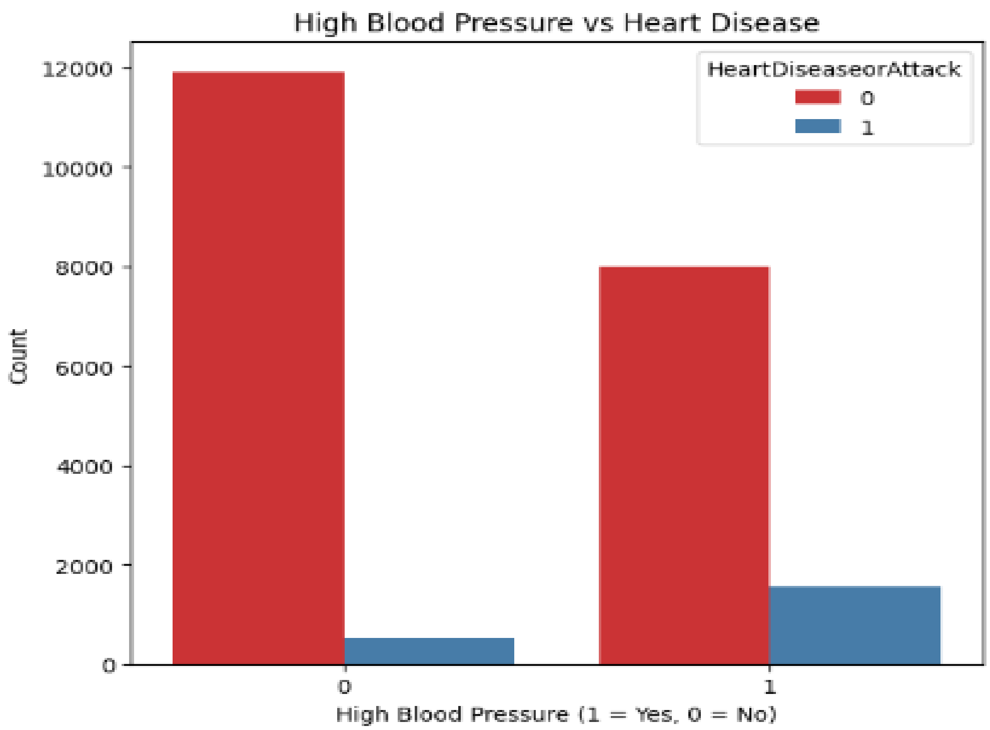
BIVARIATE ANALYSIS

```
import seaborn as sns
```

a. Explore Relationships with Heart Disease: Compare the distribution of key variables (e.g., HighBP, HighChol, BMI) between individuals with and without heart disease.

Bar Chart for High Blood Pressure vs Heart Disease

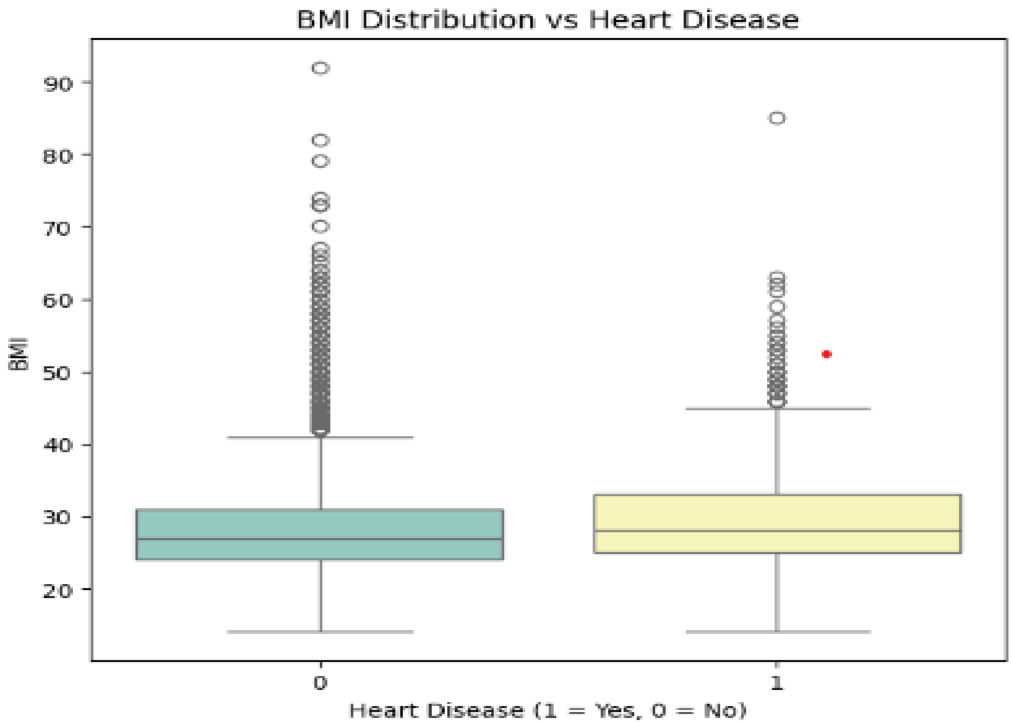
```
# Bar chart for HighBP
sns.countplot(x='HighBP', hue='HeartDiseaseorAttack', data=df, ax=axes[0], palette='Set1')
axes[0].set_title('High Blood Pressure vs Heart Disease')
axes[0].set_xlabel('High Blood Pressure (1 = Yes, 0 = No)')
axes[0].set_ylabel('Count')
```



Those with heart disease are more likely to have high blood pressure compared to those without heart disease. This suggests a strong relationship between high blood pressure and heart disease.

Boxplot for BMI vs Heart Disease

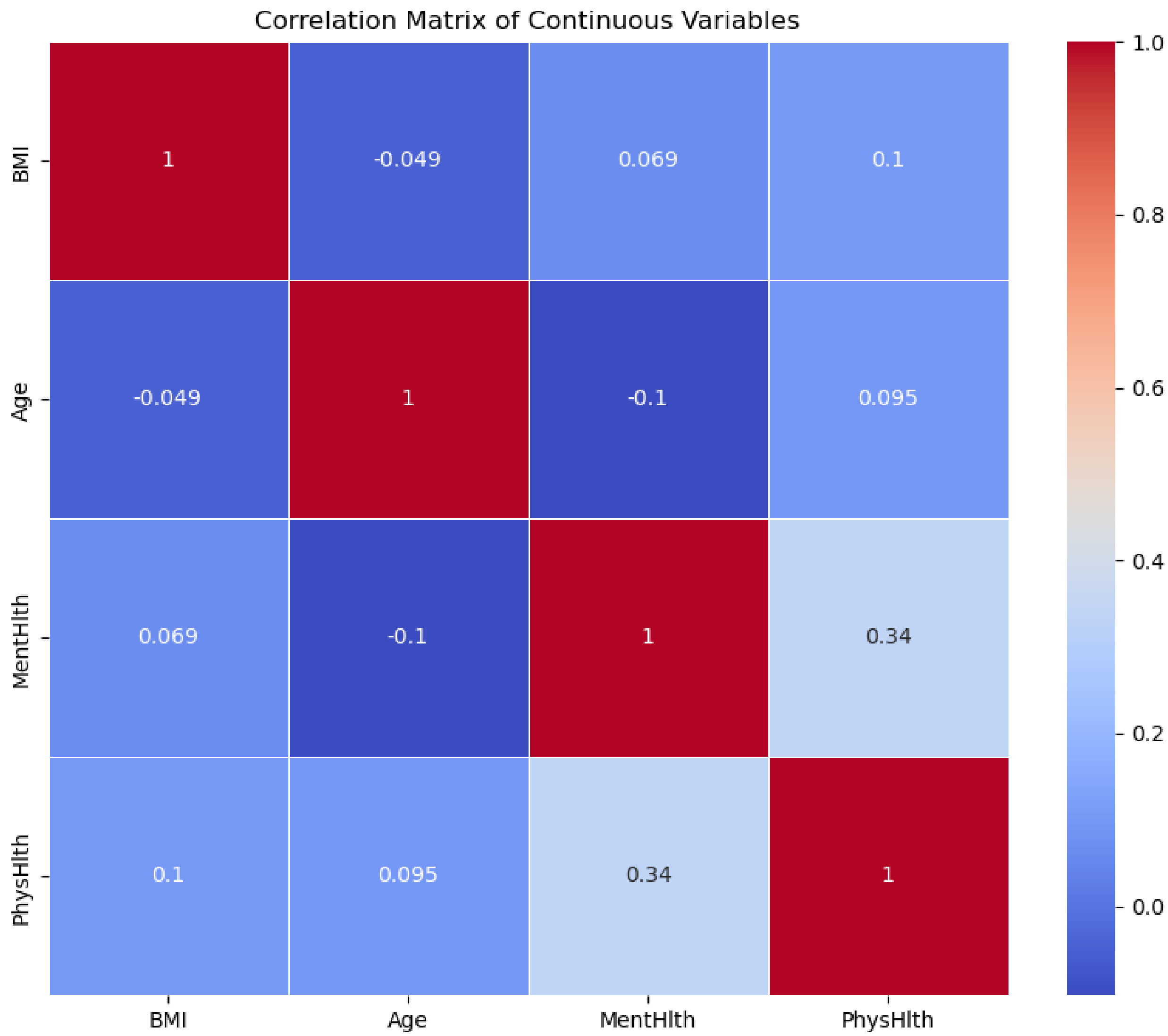
```
# Box plot for BMI
sns.boxplot(x='HeartDiseaseorAttack', y='BMI', data=df, ax=axes[2], palette='Set3')
axes[2].set_title('BMI Distribution vs Heart Disease')
axes[2].set_xlabel('Heart Disease (1 = Yes, 0 = No)')
axes[2].set_ylabel('BMI')
```



The BMI distribution for those with heart disease appears similar to those without heart disease, though individuals with heart disease might have slightly higher BMI on average.

b. Visualize Correlations Between Variables: Use scatter plots or correlation matrices to explore relationships between continuous variables.

```
plt.figure(figsize=(10, 8))
corr = df[['BMI', 'Age', 'MentHlth', 'PhysHlth']].corr()  # Select
continuous variables
sns.heatmap(corr, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Matrix of Continuous Variables')
plt.show()
```



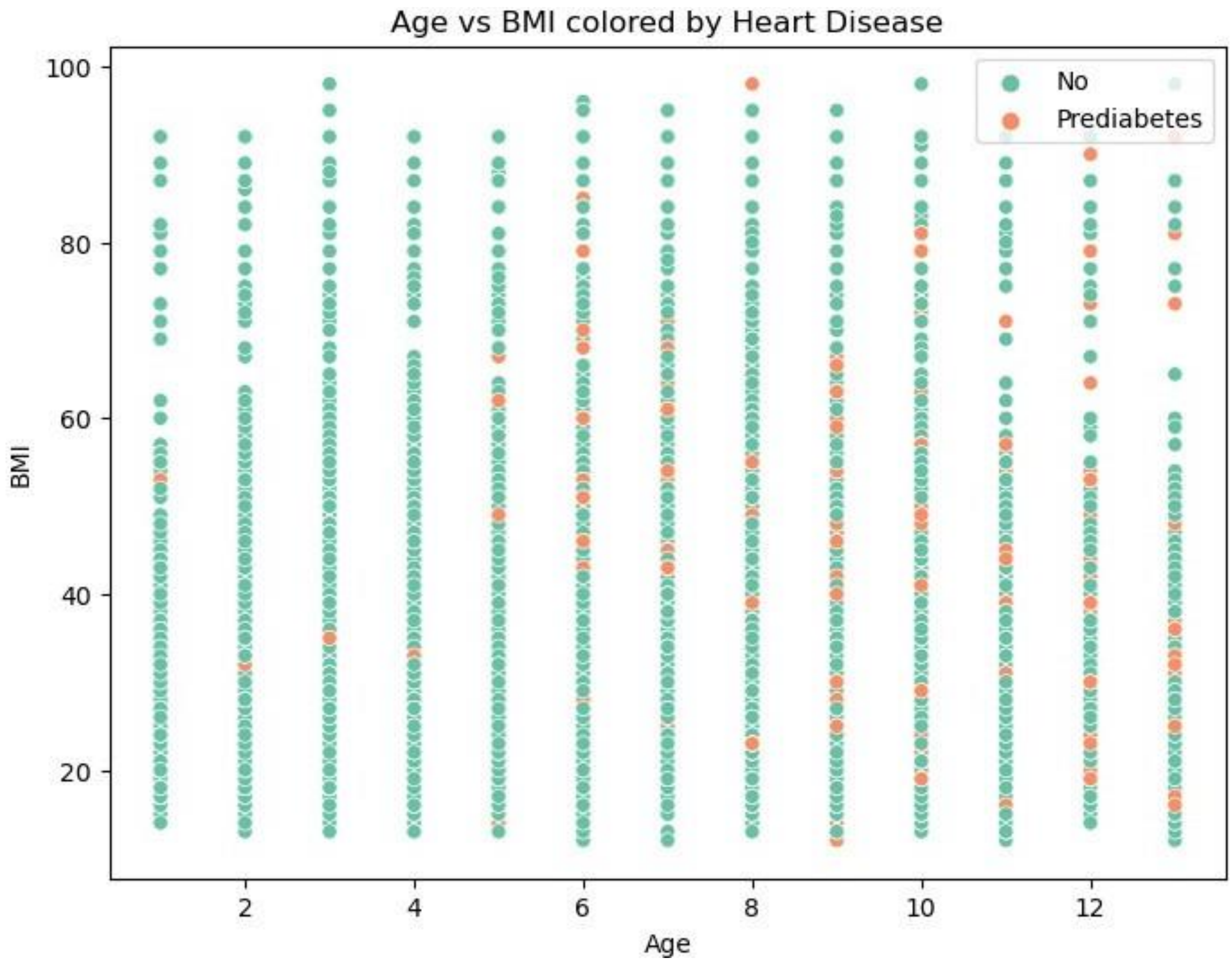
1. BMI and physical health days show a weak positive correlation (0.12).

2. Age and BMI have a slight negative correlation (-0.037), indicating that BMI tends to slightly decrease with age.

3. There is a stronger correlation (0.35) between BMI and physical health days, suggesting that individuals with higher BMI may experience more physical health issues.

```
# Scatter Plot - Age vs BMI with legend placement
plt.figure(figsize=(8, 6))
sns.scatterplot(x='Age', y='BMI', hue='HeartDiseaseorAttack', data=df,
palette='Set2')
plt.title('Age vs BMI colored by Heart Disease')
plt.xlabel('Age')
plt.ylabel('BMI')
```

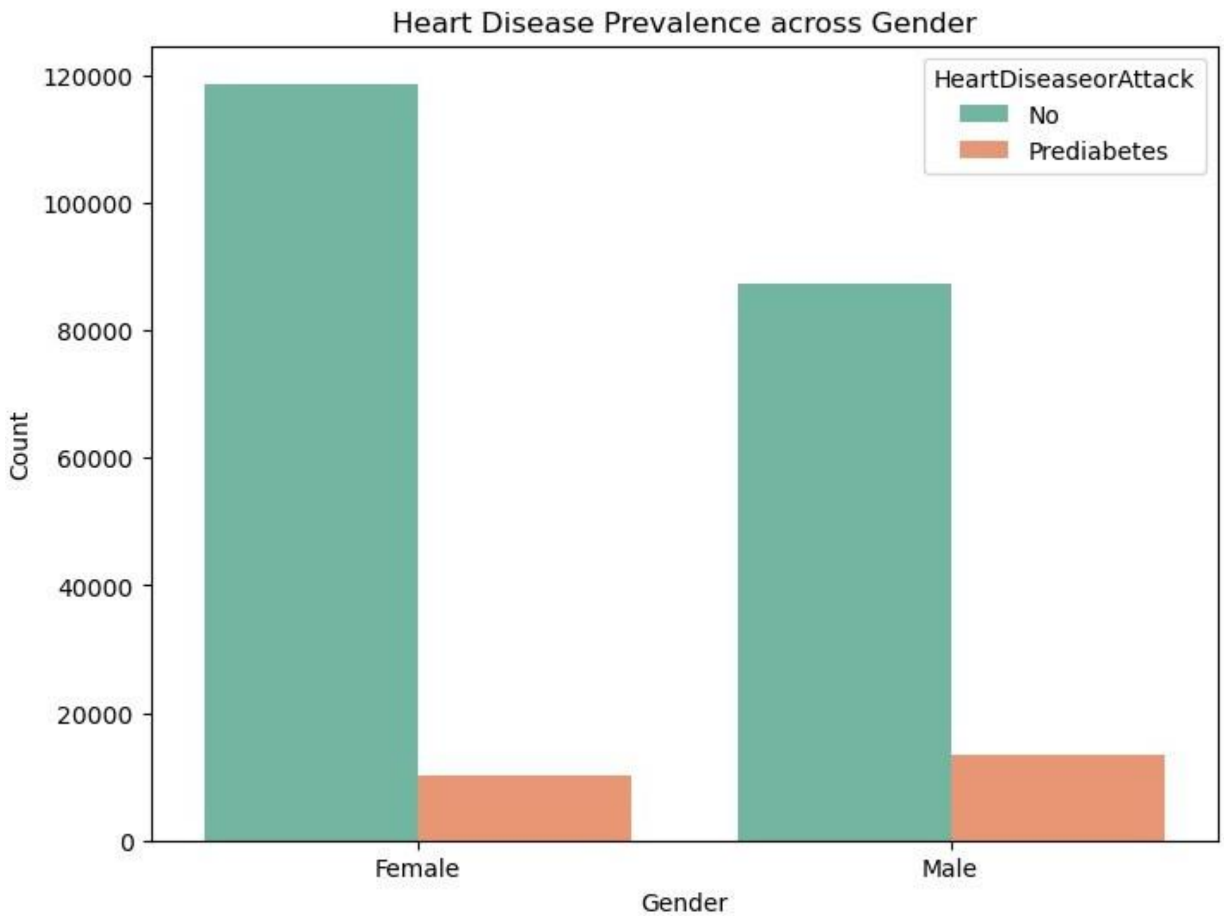
```
plt.legend(loc='upper right')  # Specify the location
plt.show()
```



c. Compare Heart Disease Across Demographic Groups: Analyze the prevalence of heart disease across different demographic groups (e.g., age, gender, education) using bar charts or pivot tables.

Bar Chart for Heart Disease across Gender

```
plt.figure(figsize=(8, 6))
sns.countplot(x='Sex', hue='HeartDiseaseorAttack', data=df,
palette='Set2')
plt.title('Heart Disease Prevalence across Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
```



The majority of individuals, irrespective of gender, do not suffer from heart disease (as indicated by the taller green bars). The orange bars, representing individuals with heart disease, are much smaller for both genders. However, it appears that one gender has a higher overall prevalence of heart disease.

Pivot Table for Heart Disease across Education Levels

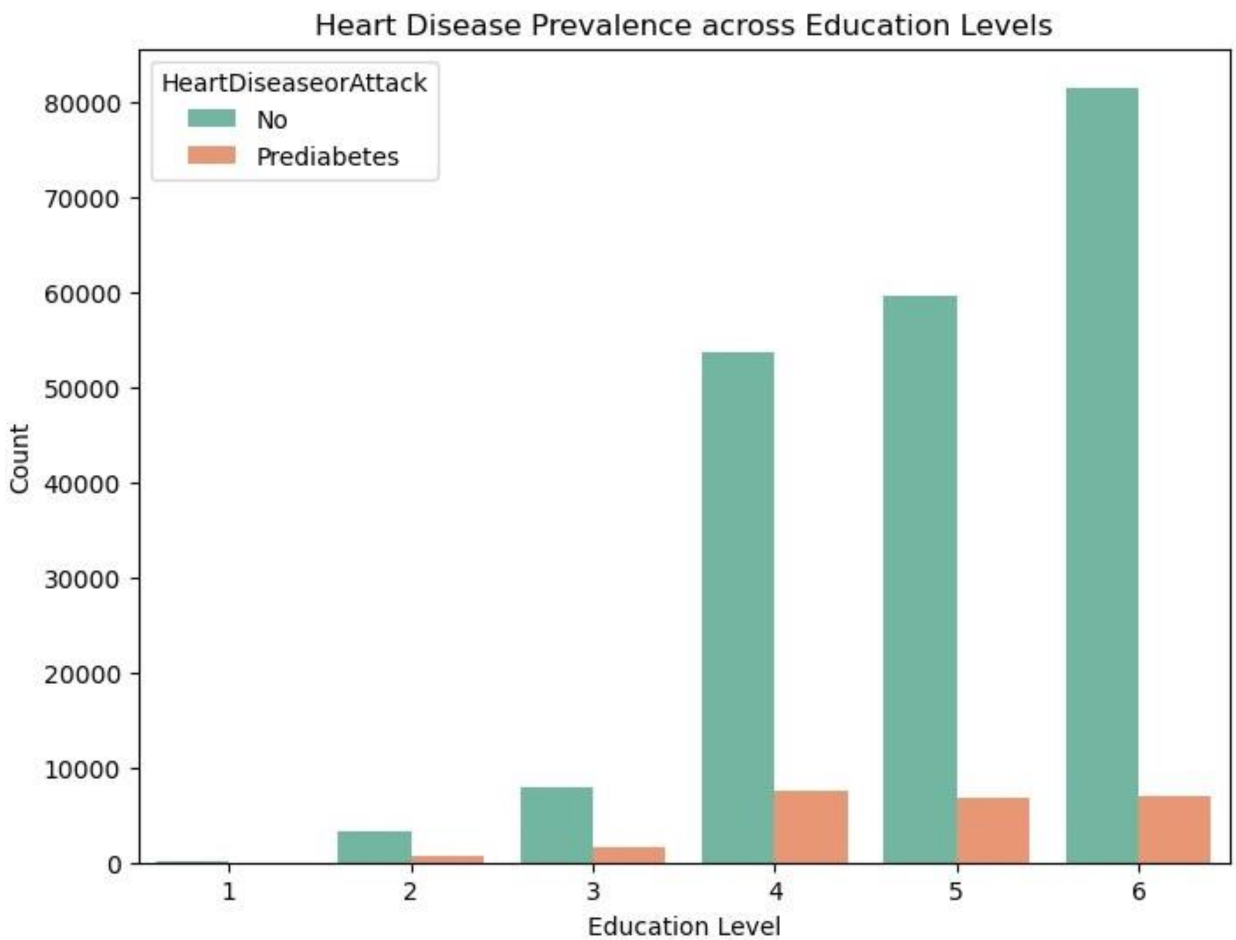
```
pivot_edu = df.pivot_table(values='HeartDiseaseorAttack',
index='Education', aggfunc='count')
print(pivot_edu)
```

```
              HeartDiseaseorAttack
Education
1                              174
2                             4040
3                             9467
4                            61158
5                            66499
6                            88443
```

As the education level increases from 1 to 6, the number of individuals increases significantly. This suggests that higher education levels have a larger representation in the dataset. The data can also be used to compare the prevalence of heart disease at different education levels by referring to the detailed count of heart disease cases.

Bar Chart for Heart Disease across Education Levels

```
plt.figure(figsize=(8, 6))
sns.countplot(x='Education', hue='HeartDiseaseorAttack', data=df,
palette='Set2')
plt.title('Heart Disease Prevalence across Education Levels')
plt.xlabel('Education Level')
plt.ylabel('Count')
plt.show()
```



As education levels increase, the overall number of individuals (both with and without heart disease) increases as well. However, the proportion of individuals with heart disease (orange bars) is relatively low at all education levels, with a slight increase in the later stages (levels 4, 5, and 6).

Heart Disease Analysis

This notebook explores various aspects of heart disease data. We will visualize categorical variables, compare heart disease prevalence across different categories, and explore relationships between variables.

1. Loading the Data

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load your dataset
df = pd.read_excel('C:/Users/Vilas/Downloads/cleaned_dataset.xlsx')

# Display the first few rows of the dataset
df.head()
```

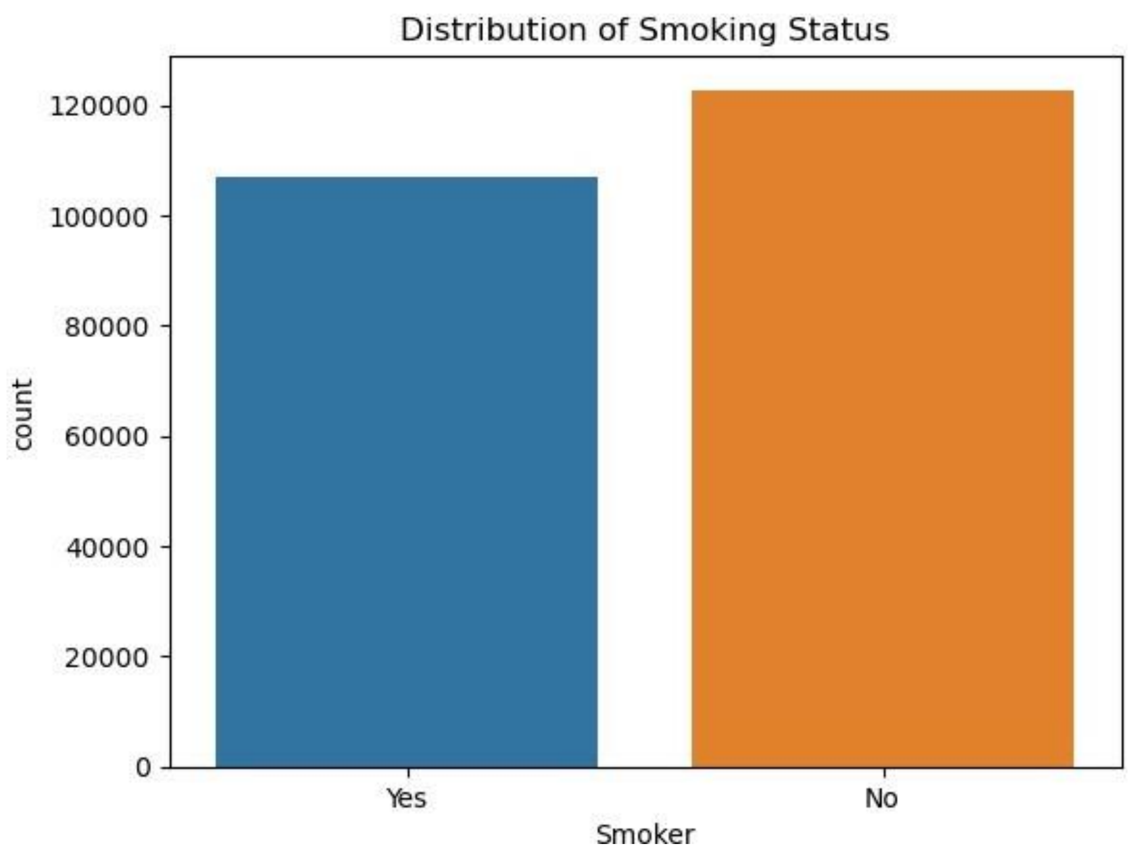
```
  HeartDiseaseorAttack HighBP  HighChol  CholCheck   BMI  Smoker  Stroke
Diabetes  \
0                    No     Yes        Yes        Yes    40     Yes      No
0
1                    No      No         No         No    25     Yes      No
0
2                    No     Yes        Yes        Yes    28      No      No
0
3                    No     Yes         No        Yes    27      No      No
0
4                    No     Yes        Yes        Yes    24      No      No
0

  PhysActivity  Fruits  ... AnyHealthcare  NoDocbcCost  GenHlth  MentHlth
\
0           No      No  ...           Yes           No        5        18

1          Yes      No  ...            No          Yes        3         0

2           No     Yes  ...           Yes          Yes        5        30

3          Yes     Yes  ...           Yes           No        2         0

4          Yes     Yes  ...           Yes           No        2         3


   PhysHlth   DiffWalk      Sex  Age  Education   Income
0        15        Yes   Female    9          4        3
1         0         No   Female    7          6        1
2        30        Yes   Female    9          4        8
```

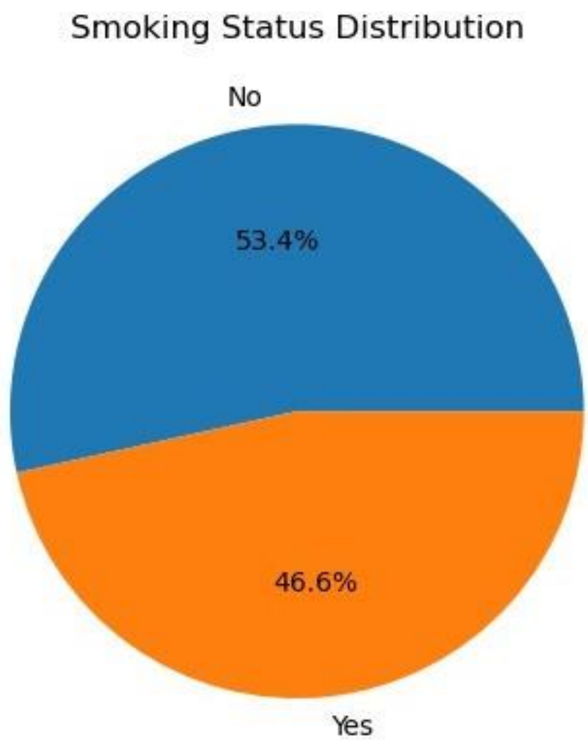
```
3           0          No   Female   11           3         6
4           0          No   Female   11           5         4

[5 rows x 22 columns]
```

2. Visualizing Categorical Variables

Smoking Status (Smoker)

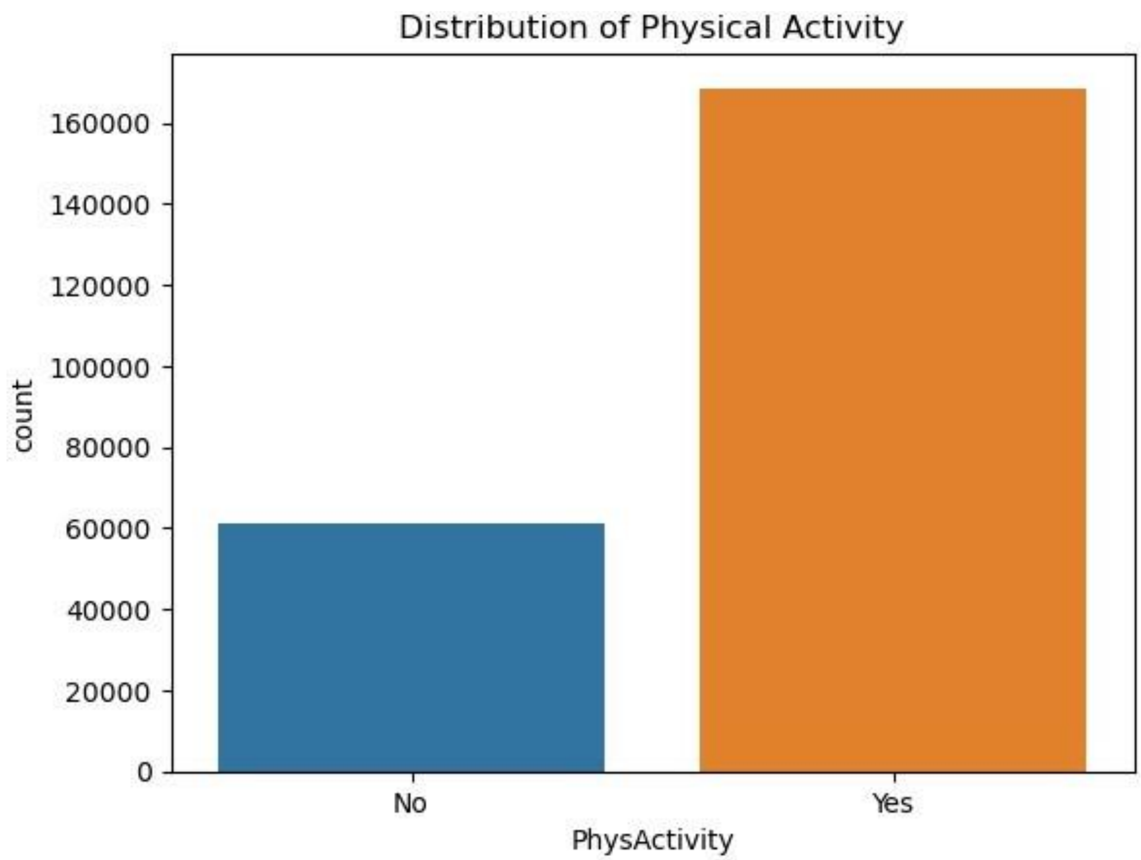
```
# Bar Chart for Smoking Status
sns.countplot(x='Smoker', data=df)
plt.title('Distribution of Smoking Status')
plt.show()

# Pie Chart for Smoking Status
smoker_counts = df['Smoker'].value_counts()
plt.pie(smoker_counts, labels=smoker_counts.index, autopct='%1.1f%%')
plt.title('Smoking Status Distribution')
plt.show()
```

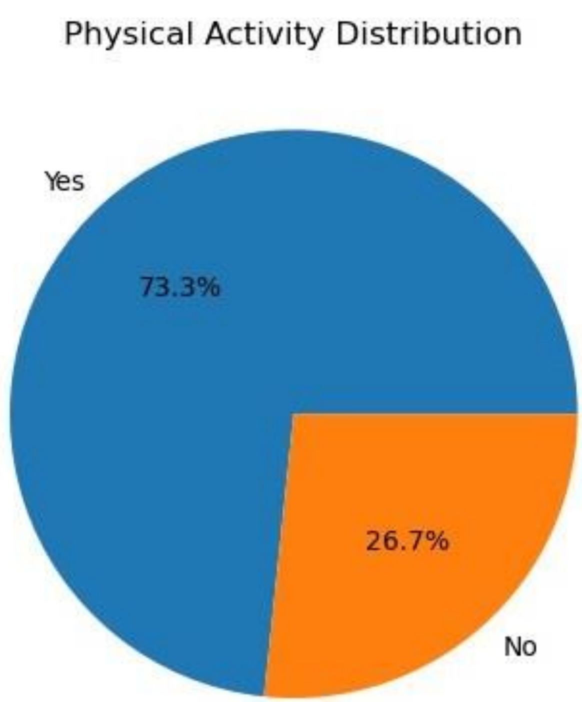




Physical Activity (PhysActivity)

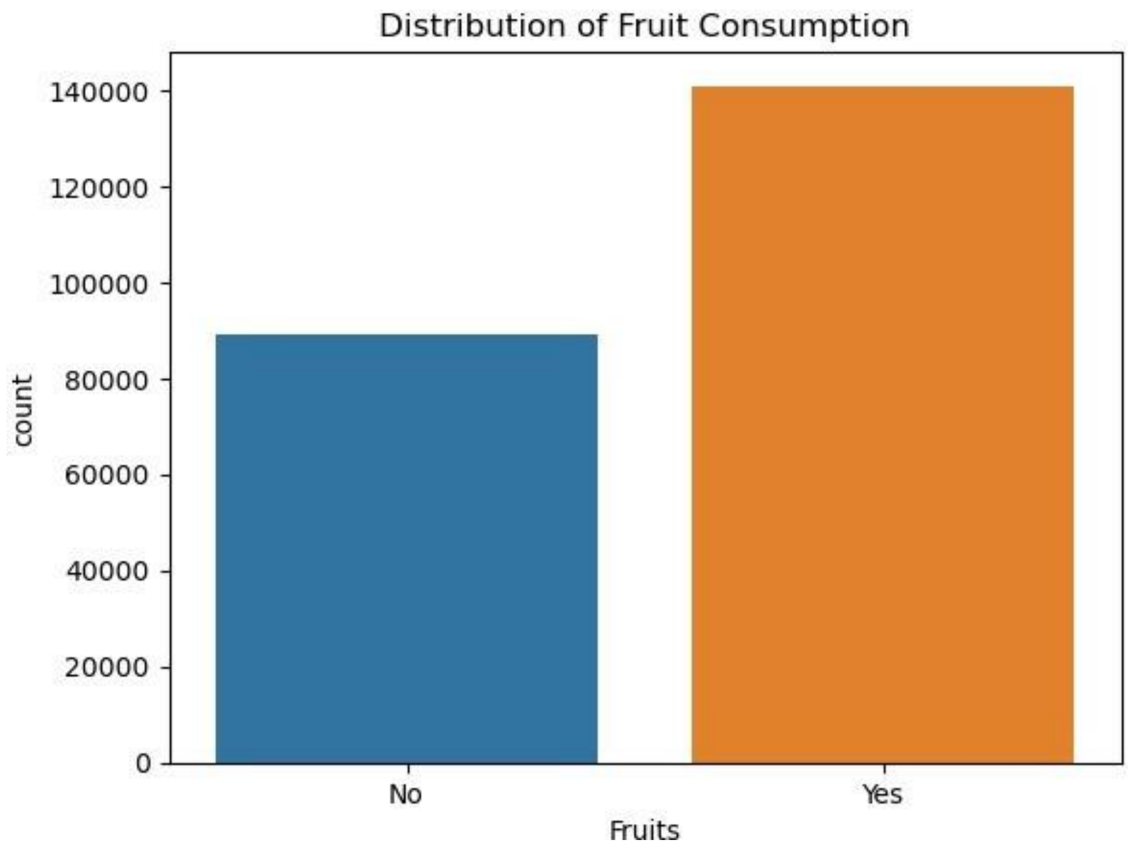
```
# Bar Chart for Physical Activity
sns.countplot(x='PhysActivity', data=df)
plt.title('Distribution of Physical Activity')
plt.show()

# Pie Chart for Physical Activity
phys_activity_counts = df['PhysActivity'].value_counts()
plt.pie(phys_activity_counts, labels=phys_activity_counts.index,
autopct='%1.1f%%')
plt.title('Physical Activity Distribution')
plt.show()
```

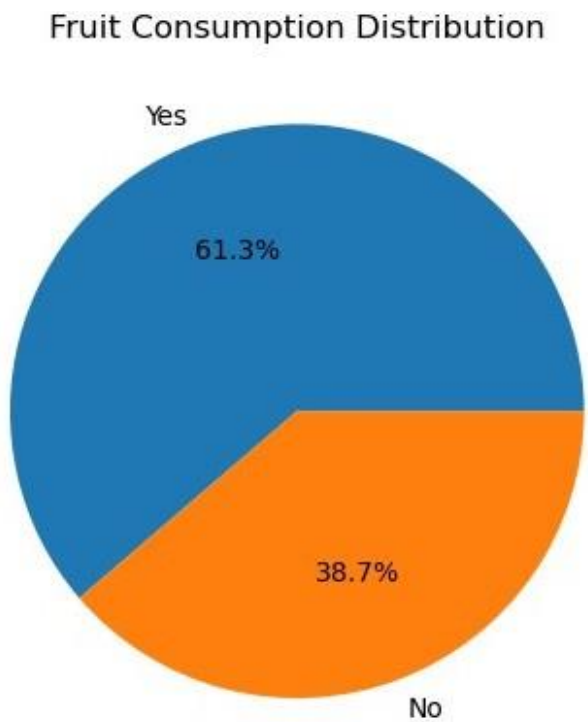

Distribution of Physical Activity
160000
140000
120000
100000
80000
60000
40000
20000
0
count
No
Yes
PhysActivity


Physical Activity Distribution
Yes
73.3%
26.7%
No

Fruit Consumption (Fruits)

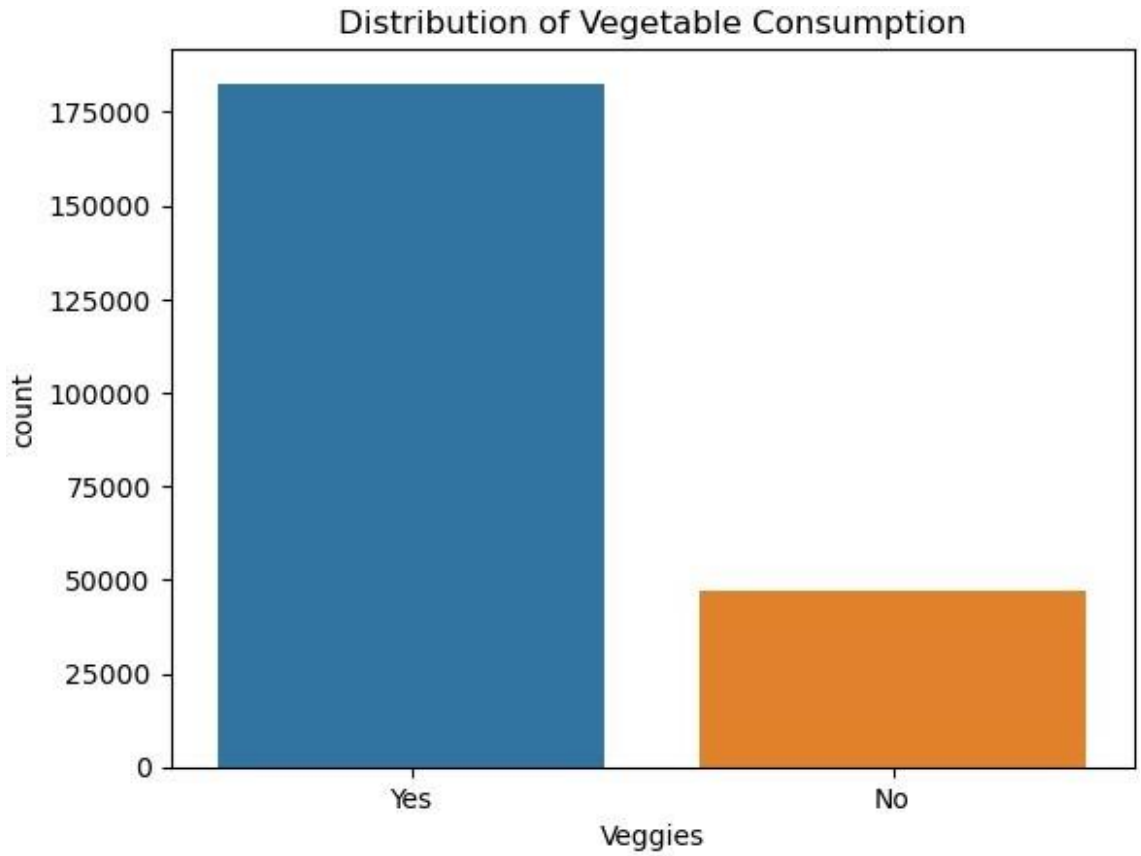
```
# Bar Chart for Fruit Consumption
sns.countplot(x='Fruits', data=df)
plt.title('Distribution of Fruit Consumption')
plt.show()

# Pie Chart for Fruit Consumption
fruit_counts = df['Fruits'].value_counts()
plt.pie(fruit_counts, labels=fruit_counts.index, autopct='%1.1f%%')
plt.title('Fruit Consumption Distribution')
plt.show()
```

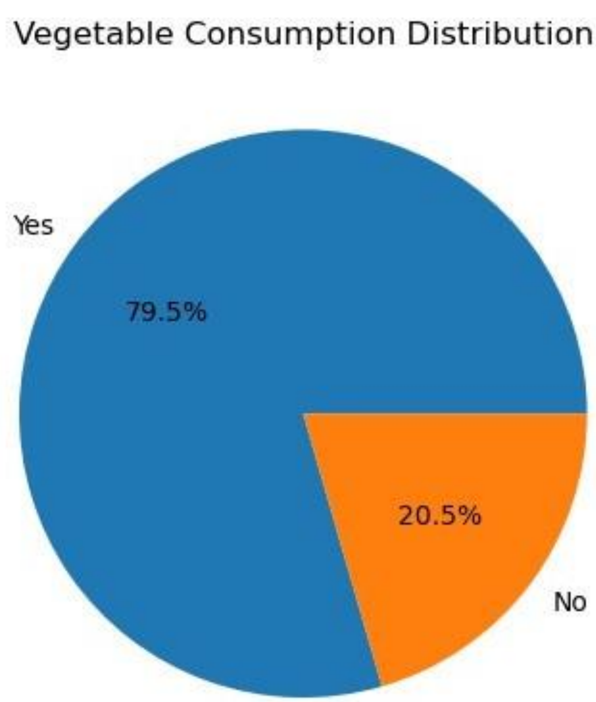




Vegetable Consumption (Veggies)

```
# Bar Chart for Vegetable Consumption
sns.countplot(x='Veggies', data=df)
plt.title('Distribution of Vegetable Consumption')
plt.show()

# Pie Chart for Vegetable Consumption
veggie_counts = df['Veggies'].value_counts()
plt.pie(veggie_counts, labels=veggie_counts.index, autopct='%1.1f%%')
plt.title('Vegetable Consumption Distribution')
plt.show()
```

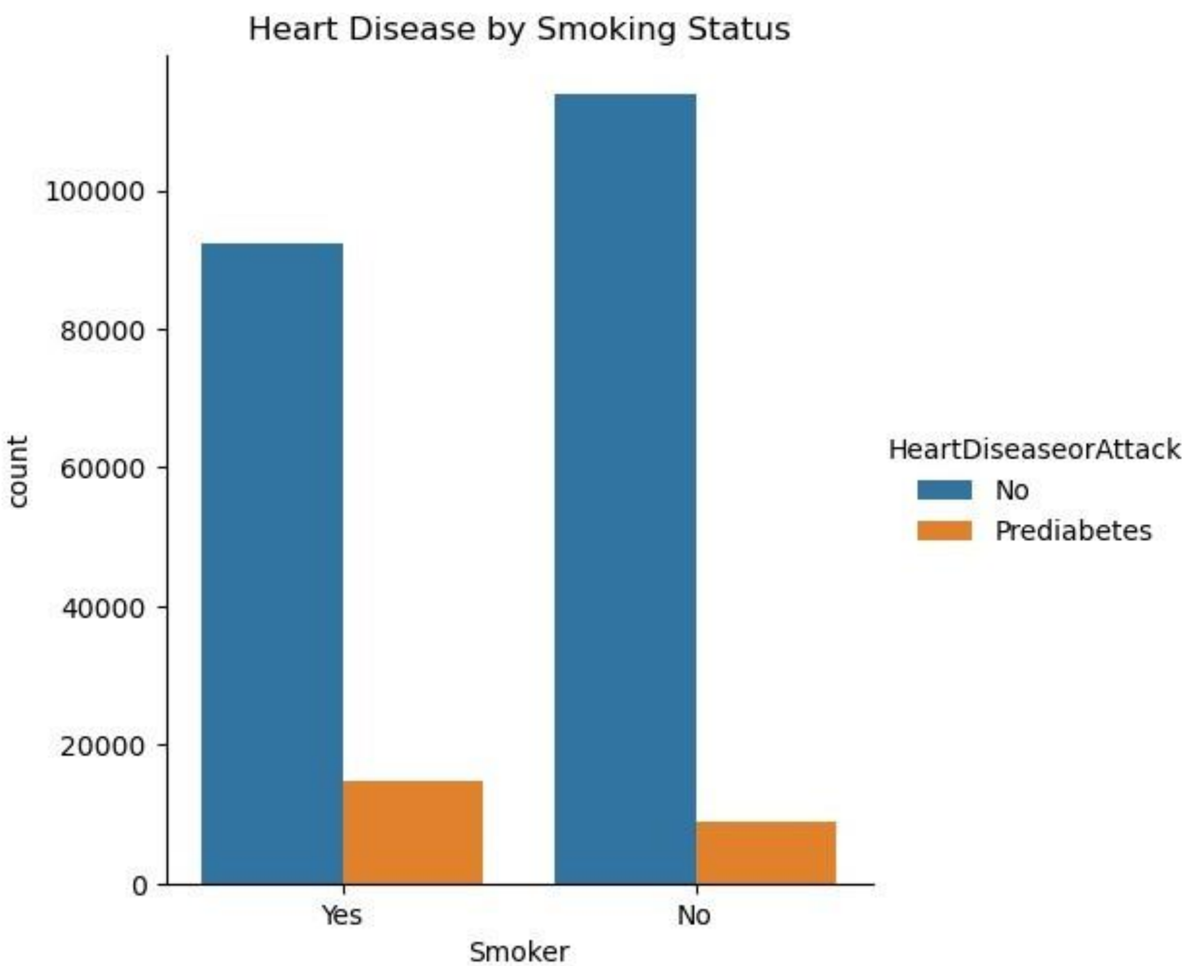

Distribution of Vegetable Consumption
175000
150000
125000
100000
75000
50000
25000
0
count
Yes
No
Veggies


Vegetable Consumption Distribution
Yes
79.5%
20.5%
No

3. Comparing Heart Disease Across Categories

Heart Disease vs. Smoking Status

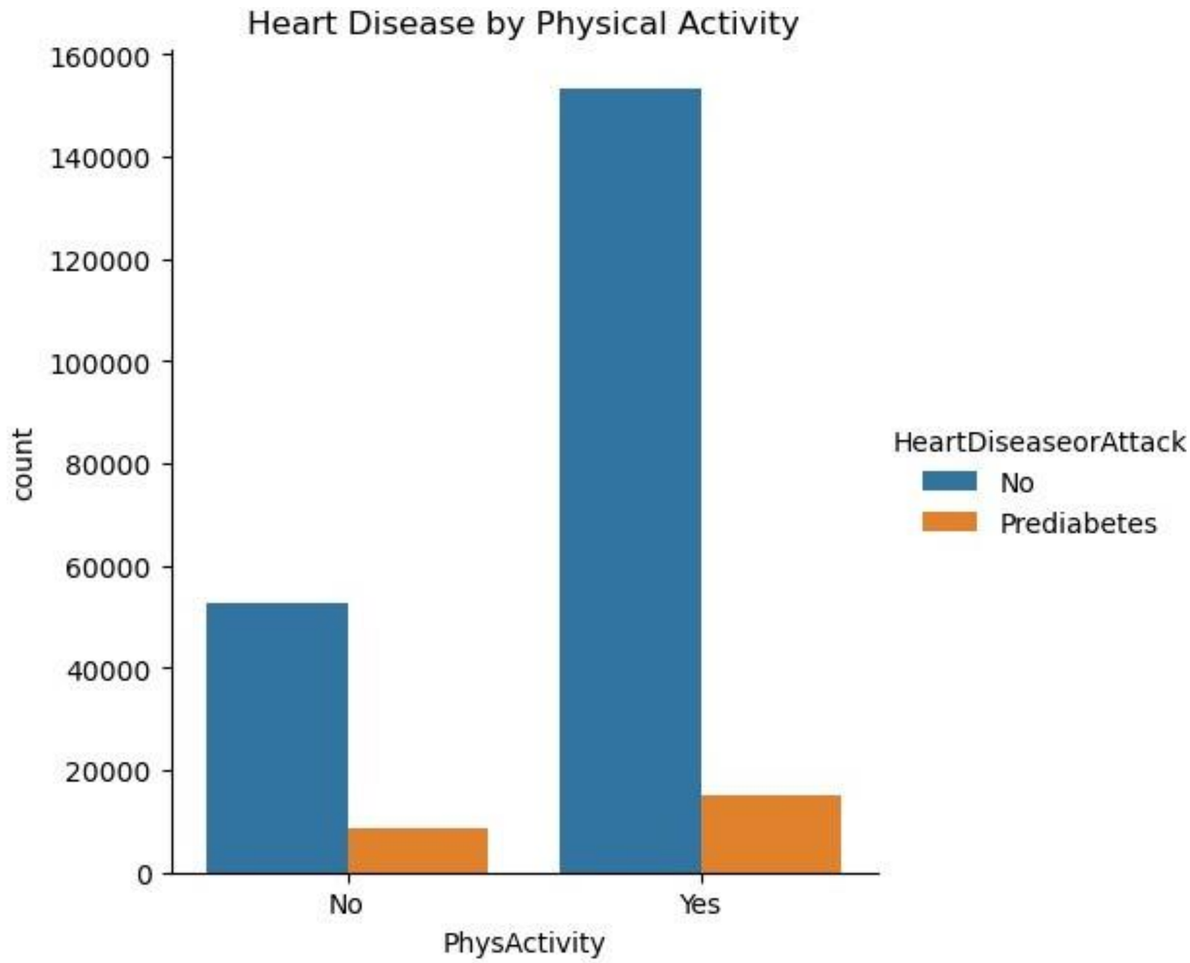
```
# Grouped Bar Chart for Heart Disease vs Smoking Status
sns.catplot(x='Smoker', hue='HeartDiseaseorAttack', kind='count',
data=df)
plt.title('Heart Disease by Smoking Status')
plt.show()

# Stacked Bar Chart (requires manual stacking)
```



Heart Disease vs. Physical Activity

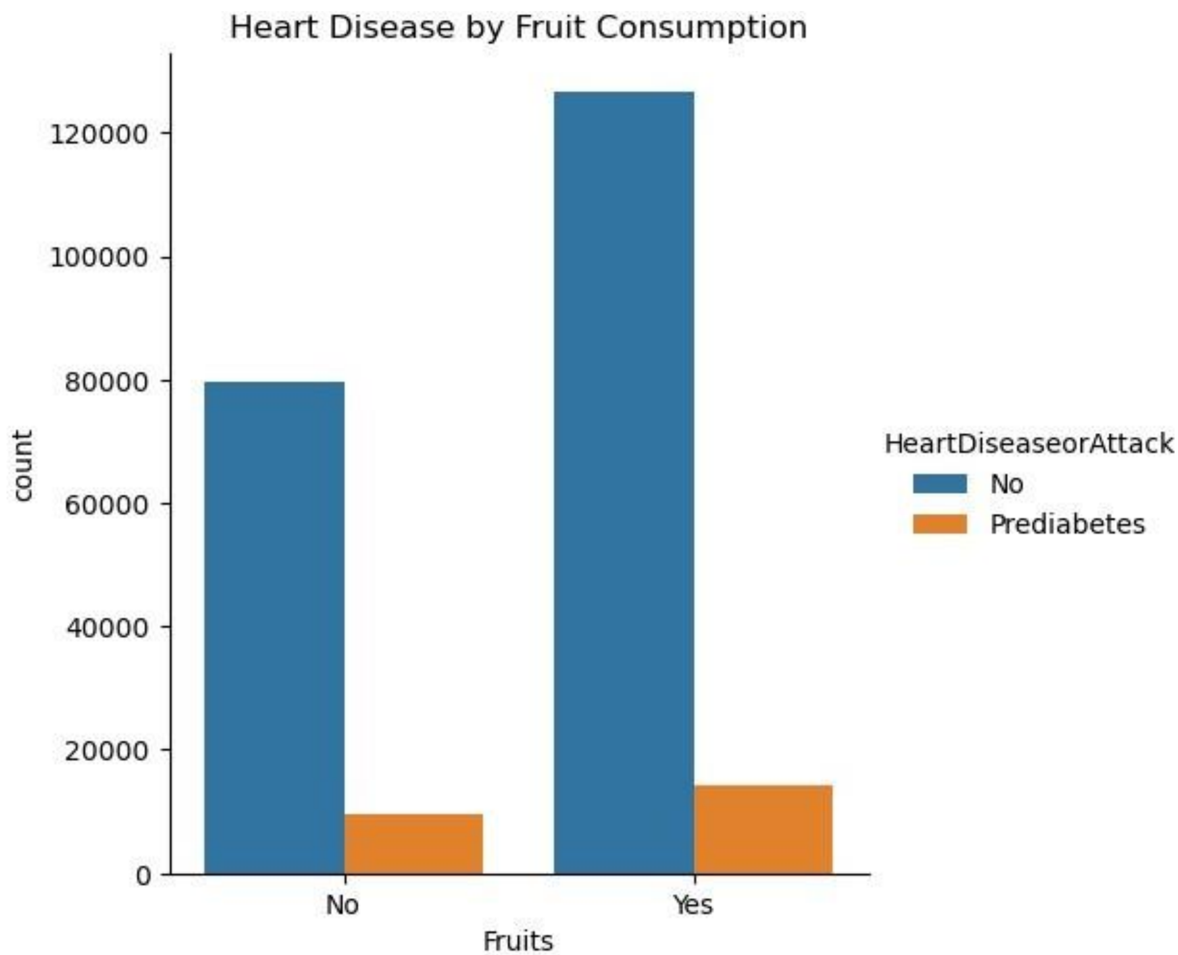
```
# Grouped Bar Chart for Heart Disease vs Physical Activity
sns.catplot(x='PhysActivity', hue='HeartDiseaseorAttack',
kind='count', data=df)
plt.title('Heart Disease by Physical Activity')
plt.show()
```

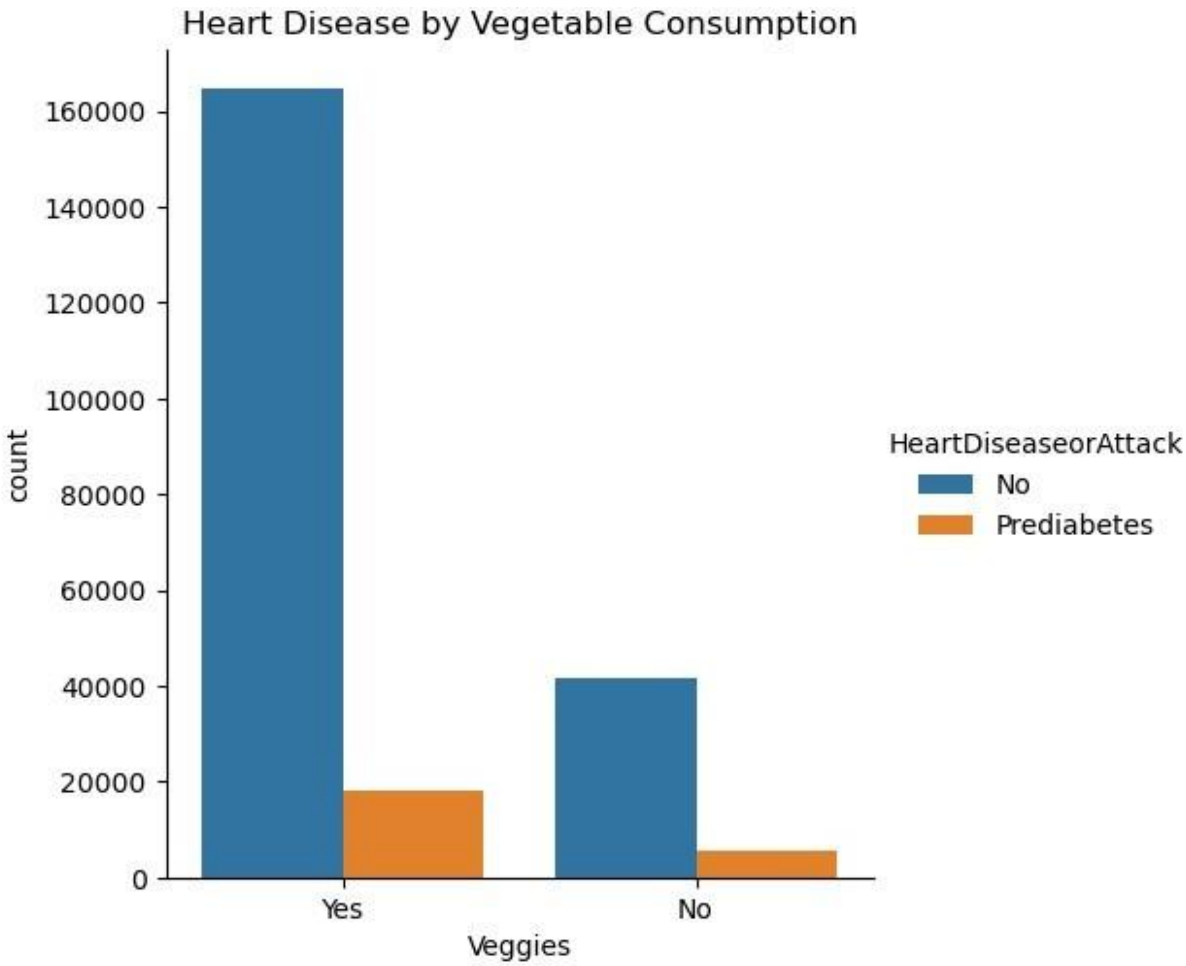
```
# Stacked Bar Chart (requires manual stacking)
```



Heart Disease vs. Fruit and Vegetable Consumption

```
# Grouped Bar Chart for Heart Disease vs Fruit Consumption
sns.catplot(x='Fruits', hue='HeartDiseaseorAttack', kind='count',
data=df)
plt.title('Heart Disease by Fruit Consumption')
plt.show()

# Grouped Bar Chart for Heart Disease vs Vegetable Consumption
sns.catplot(x='Veggies', hue='HeartDiseaseorAttack', kind='count',
data=df)
plt.title('Heart Disease by Vegetable Consumption')
plt.show()
```

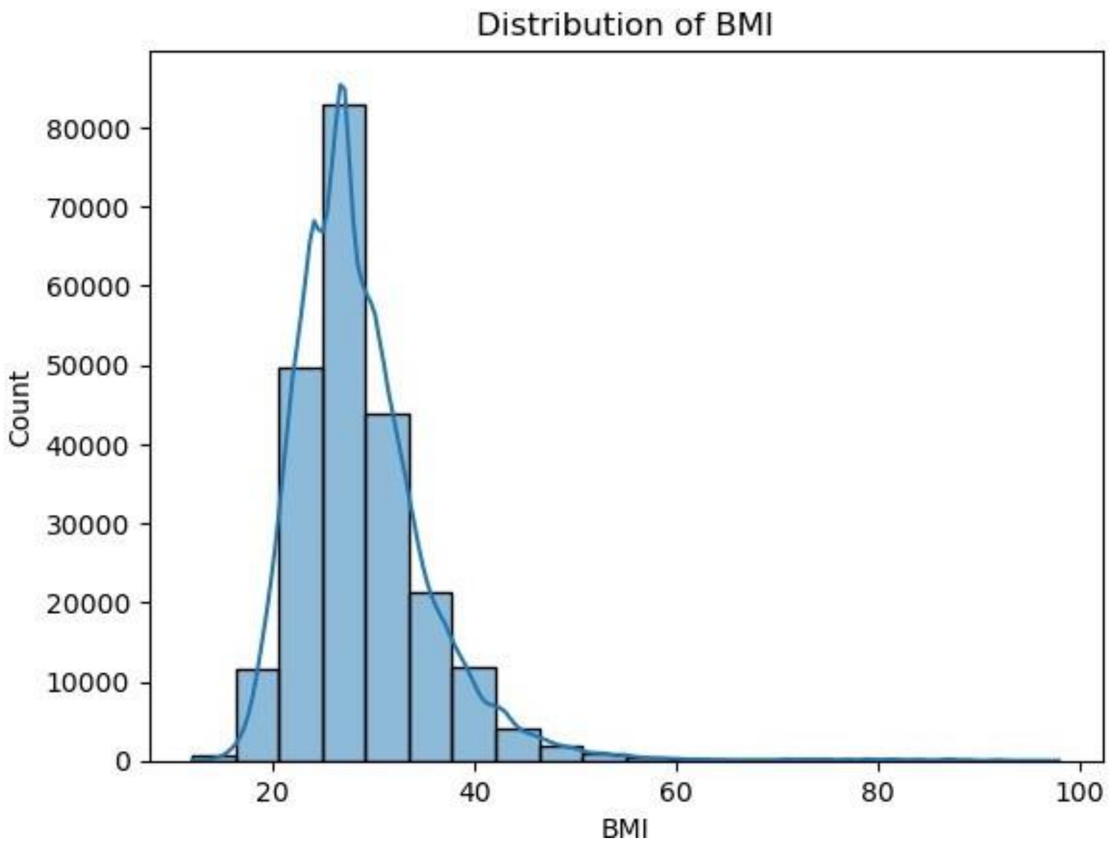

Heart Disease by Fruit Consumption
count
0
20000
40000
60000
80000
100000
120000
No
Yes
Fruits
HeartDiseaseorAttack
No
Prediabetes

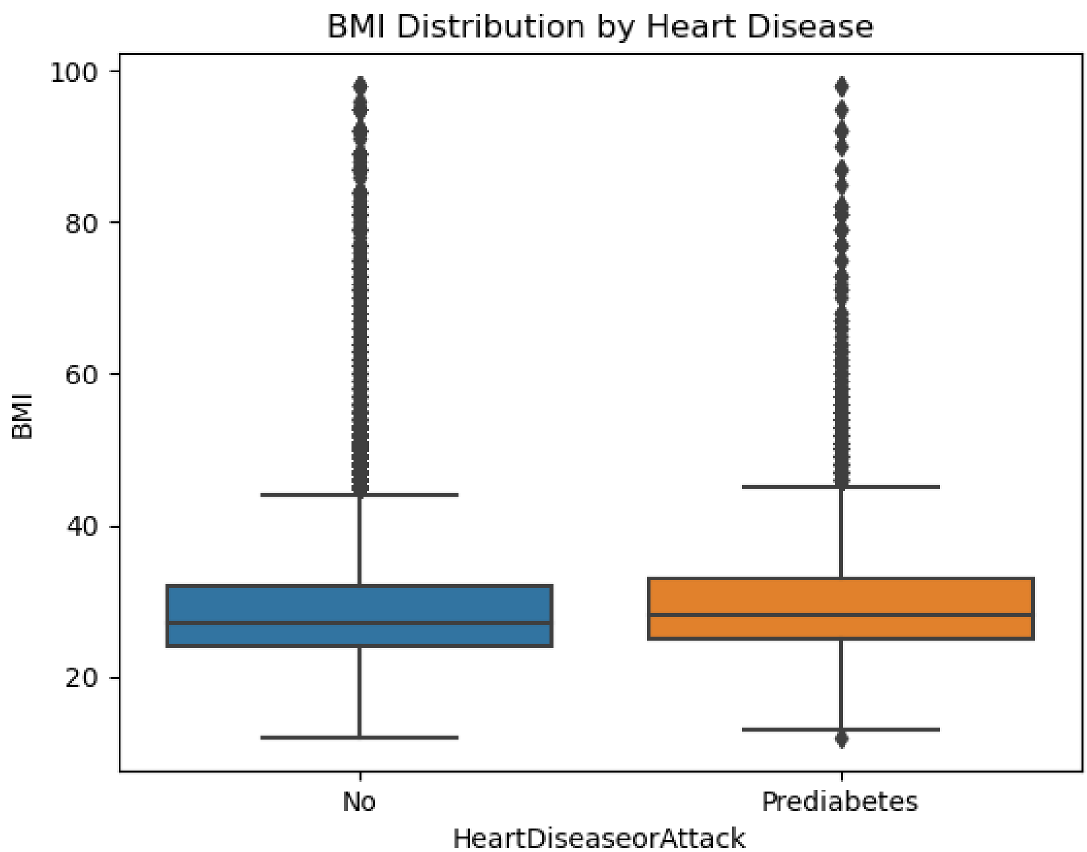


Exploring Other Visualization Techniques

Distributions of Continuous Variables

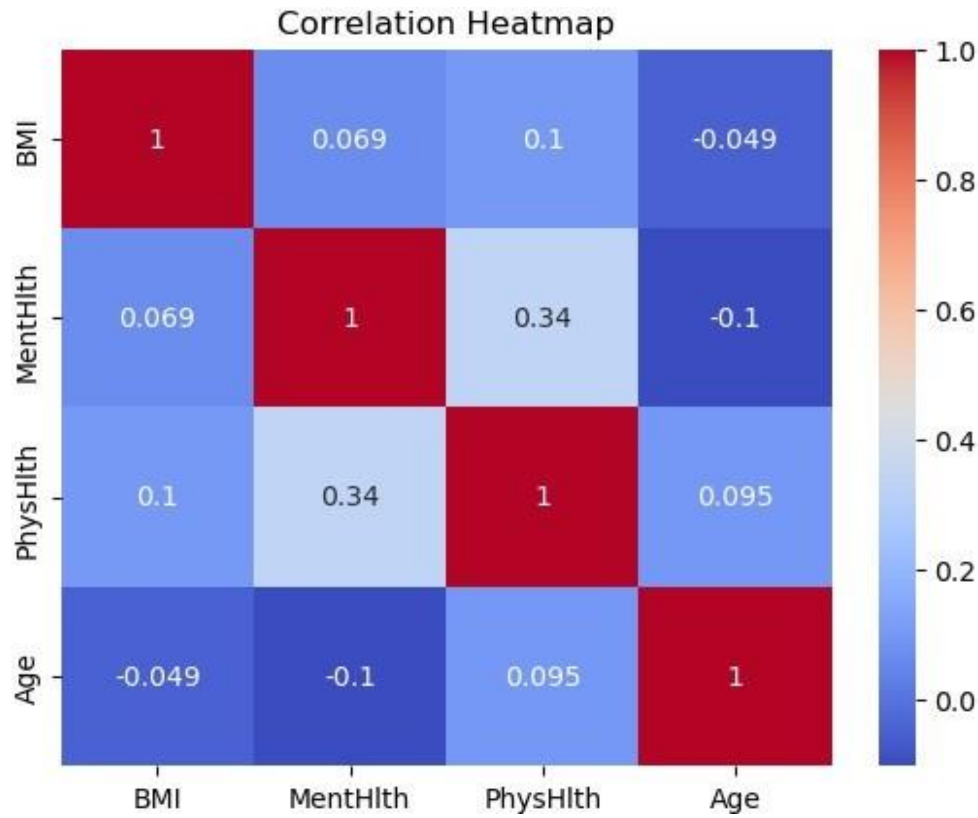
```
# Histogram for BMI
sns.histplot(df['BMI'], bins=20, kde=True)
plt.title('Distribution of BMI')
plt.show()

# Box Plot for BMI by Heart Disease
sns.boxplot(x='HeartDiseaseorAttack', y='BMI', data=df)
plt.title('BMI Distribution by Heart Disease')
plt.show()
```


Distribution of BMI
Count
80000
70000
60000
50000
40000
30000
20000
10000
0
20
40
60
80
100
BMI



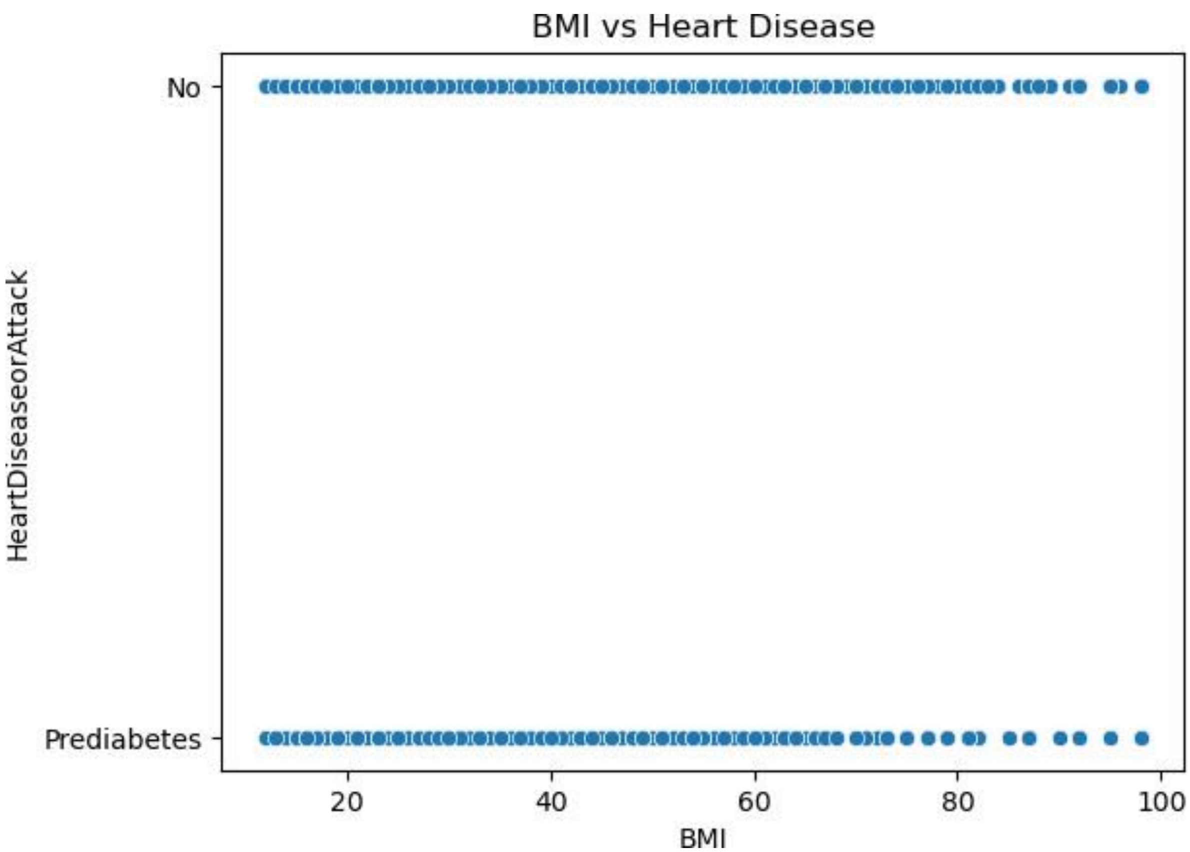
Heatmaps and Correlation Plots

```
# Correlation Heatmap
corr_matrix = df[['BMI', 'MentHlth', 'PhysHlth', 'Age']].corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



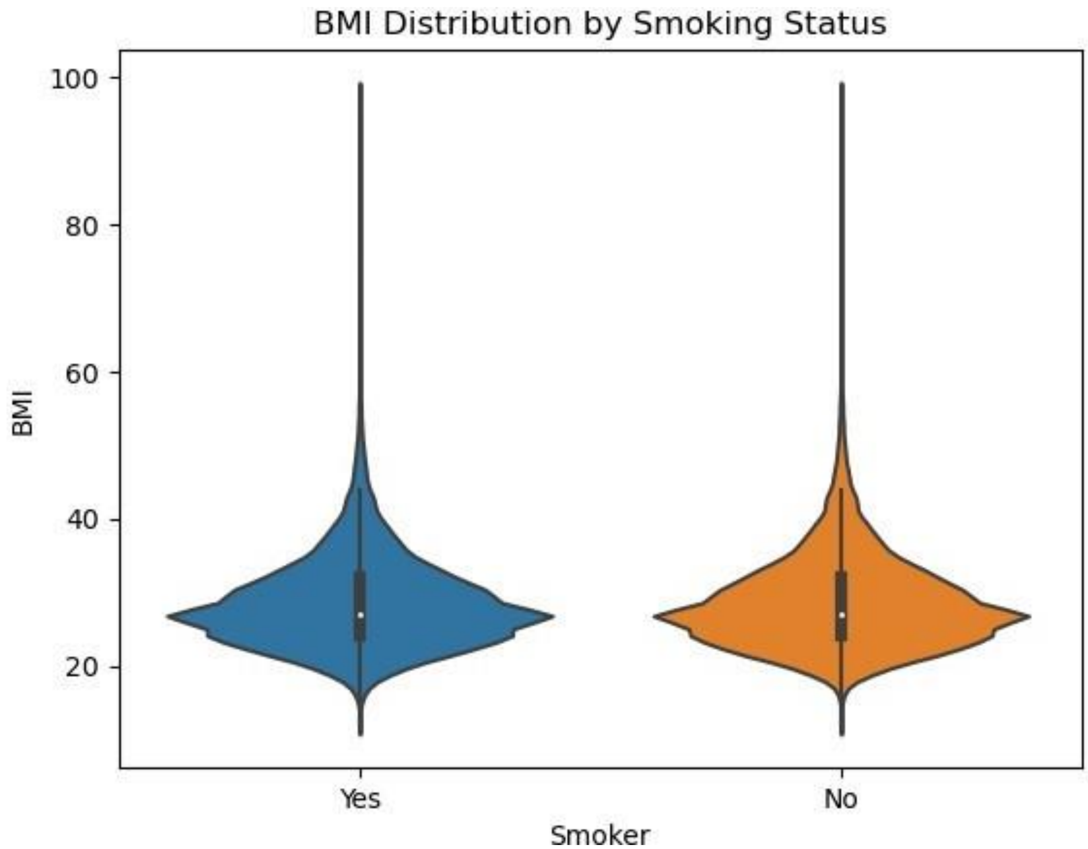
Scatter Plots

```
# Scatter Plot for BMI vs Heart Disease
sns.scatterplot(x='BMI', y='HeartDiseaseorAttack', data=df)
plt.title('BMI vs Heart Disease')
plt.show()
```



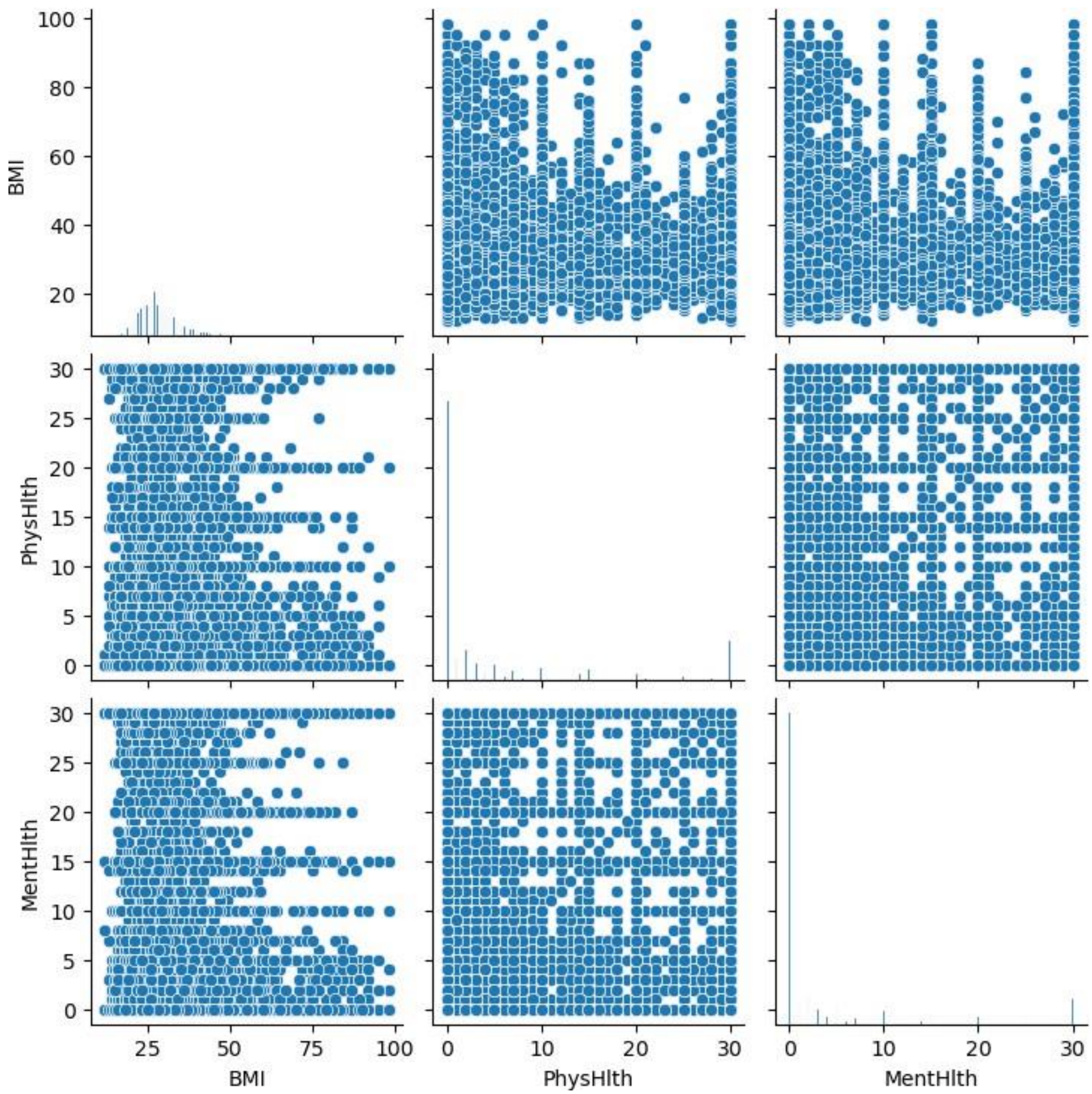
Violin Plots

```
# Violin Plot for BMI by Smoking Status
sns.violinplot(x='Smoker', y='BMI', data=df)
plt.title('BMI Distribution by Smoking Status')
plt.show()
```



Pair Plots

```
# Pair Plot for Multiple Variables
sns.pairplot(df[['BMI', 'PhysHlth', 'MentHlth',
'HeartDiseaseorAttack']])
plt.show()
```


BMI
PhysHlth
MentHlth
BMI
PhysHlth
MentHlth

Future Scope

Based on the findings regarding the factors associated with heart disease, the following recommendations and insights can be drawn for further research, interventions, and health strategies:

1. Implement Community Health Programs

Focus on Education: Develop community-based education programs emphasizing the importance of regular health screenings for high blood pressure and cholesterol levels. Highlight the risk factors of smoking, obesity, and inactivity.

Promote Healthy Lifestyles: Create initiatives to encourage physical activity and healthy eating. This could include local exercise programs, nutritional workshops, and cooking classes focused on heart-healthy diets rich in fruits, vegetables, and whole grains.

2. Enhance Screening and Early Detection

Regular Health Screenings: Encourage regular health check-ups for early detection of risk factors such as high blood pressure, high cholesterol, and diabetes, especially in high-risk populations (e.g., older adults, individuals with family histories of heart disease).

Targeted Screenings: Develop targeted screening programs for individuals with multiple risk factors or those who are overweight or obese.

3. Policy Advocacy for Smoking Cessation

Support Smoking Cessation Programs: Advocate for increased funding and support for smoking cessation programs. Implement policies that restrict smoking in public places and reduce tobacco marketing.

Incentives for Quitting: Provide incentives for individuals who quit smoking, such as reduced health insurance premiums or free counseling services.

4. Nutrition and Diet Interventions

Dietary Counseling: Offer dietary counseling and support to help individuals make healthier food choices. This may include meal planning assistance and guidance on managing portion sizes.

Access to Healthy Foods: Work with local governments to improve access to healthy foods in underserved communities, possibly through farmers' markets or community gardens.

5. Physical Activity Promotion

Create Safe Spaces: Develop and maintain parks, walking trails, and recreational facilities to promote physical activity. Organize community events such as fun runs, walks, or fitness challenges.

Incorporate Exercise into Daily Life: Encourage individuals to integrate physical activity into their daily routines, such as walking or biking to work, taking the stairs, or engaging in family sports.

6. Personalized Health Interventions
Risk Assessment Tools: Develop tools for healthcare providers to assess individual heart disease risk based on personal health data and lifestyle choices. This can help tailor interventions and recommendations to individuals.
Use of Technology: Leverage mobile health technologies (apps, wearables) to monitor health metrics (e.g., heart rate, activity levels, dietary habits) and provide personalized feedback and encouragement.

7. Further Research Opportunities
Longitudinal Studies: Conduct longitudinal studies to track the impact of lifestyle changes on heart disease prevalence and management over time.
Examine Social Determinants: Investigate how socioeconomic factors, access to healthcare, and educational background affect heart disease risk in various populations.
Cultural Influences: Explore the cultural influences on dietary habits and physical activity levels to design more culturally competent interventions.

8. Mental Health Considerations
Integrate Mental Health Services: Recognize the link between mental health and heart disease. Integrate mental health services into primary care, especially for individuals dealing with chronic illnesses, stress, or depression.
Support Systems: Create support systems for individuals struggling with lifestyle changes, including peer support groups and counseling services.

By addressing the modifiable risk factors identified in the EDA and implementing targeted interventions, communities can reduce the incidence of heart disease. A multi-faceted approach involving education, policy changes, and community support will be crucial for improving heart health outcomes and enhancing the overall well-being of populations at risk.

Limitations

The analysis of the "Heart Disease Health Indicators Dataset" has several limitations. The dataset may not include all relevant health indicators, potentially omitting critical risk factors. Its cross-sectional nature restricts the establishment of causal relationships. Many variables are self-reported, introducing bias due to inaccuracies. The findings may not be generalizable to broader populations, and the absence of longitudinal data limits assessments of changes over time. Additionally, unmeasured confounding variables could influence heart disease risk, and variability in data quality may impact the robustness and reliability of the analysis. Recognizing these limitations is essential for accurate interpretation.

Conclusion

Based on the exploratory data analysis (EDA), several factors are strongly associated with heart disease. Below is a discussion of the most significant factors:

1. High Blood Pressure (Hypertension)
Prevalence: High blood pressure is significantly more prevalent among individuals with heart disease (75.03%) compared to those without (39.56%).
Impact: The strong statistical association between high blood pressure and heart disease (confirmed by the chi-square test) suggests that individuals with hypertension are at a substantially higher risk of developing heart disease. This association highlights hypertension as a primary risk factor.

2. High Cholesterol
Prevalence: High cholesterol is more common in individuals with heart disease compared to those without. Around 16% of individuals with high cholesterol have heart disease.
Impact: The analysis shows a strong association (p-value = 0.0) between high cholesterol levels and heart disease, suggesting that cholesterol management is crucial for heart disease prevention.

3. Smoking
Prevalence: Smoking is notably more common among individuals with heart disease (13%) compared to non-smokers (6%).
Impact: Smoking has long been recognized as a major cardiovascular risk factor, and the analysis reinforces this by showing a significant association between smoking and heart disease risk.

4. Stroke
Prevalence: Individuals who have suffered a stroke are at a greater risk for heart disease (38%) compared to those without a stroke history (8%).
Impact: Stroke, being a cerebrovascular event, shares many risk factors with heart disease. The overlap between these conditions in the dataset highlights stroke as a major predictor of future heart disease.

5. Body Mass Index (BMI)
Prevalence: Higher BMI values are strongly correlated with heart disease. Obese individuals (BMI ≥ 30) have a higher prevalence of heart disease (11.5%) compared to those with normal weight (6.8%).
Impact: Obesity is associated with increased cardiovascular strain, hypertension, and high cholesterol, which are all contributors to heart disease. The low p-value for BMI in the analysis confirms it as a significant risk factor.

6. Diabetes (Type 2)
Prevalence: Individuals with diabetes, particularly type 2, have a much higher prevalence of heart disease (22%) compared to non-diabetics (7%).
Impact: Diabetes, especially type 2, is associated with numerous metabolic complications that contribute to heart disease. The analysis confirms diabetes as a key predictor of cardiovascular problems.

7. Physical Inactivity
Prevalence: Physically inactive individuals show a higher prevalence of heart disease (14%) compared to those who engage in regular physical activity (8%).
Impact: The protective effect of physical activity against heart disease is evident, suggesting that promoting an active lifestyle is vital for heart disease prevention.

8. Age
Prevalence: Heart disease prevalence increases significantly with age, reaching 24% in the oldest age group.
Impact: Age is a well-known, non-modifiable risk factor for heart disease, with older individuals being at higher risk due to the cumulative effects of other risk factors.

9. Sex
Prevalence: Males have a higher prevalence of heart disease (12%) compared to females (7%).
Impact: Sex differences in heart disease prevalence point to biological and lifestyle factors that may increase risk in men compared to women.

The factors most strongly associated with heart disease in the dataset include high blood pressure, high cholesterol, smoking, stroke, high BMI, diabetes, physical inactivity, age, and sex. These findings reinforce the well-established understanding that heart disease is a multifactorial condition, with both modifiable (e.g., smoking, cholesterol, hypertension) and non-modifiable (e.g., age, sex) risk factors playing significant roles.

These associations emphasize the importance of lifestyle interventions and regular health monitoring to reduce the risk of heart disease, particularly for individuals with multiple risk factors.